



Batterjee Medical College
For Science and Technology

Approach to a case of palpitation and/or Syncope

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Designation: Associate professor

Department: Medicine



Cardiac Arrhythmia

Learning Objectives

Knowledge & understanding

Describe the clinical features and ECG characteristics of the major bradyarrhythmia and tachyarrhythmias

Identify the components of the CHA2DS2-VASc risk stratification and its role in guiding anticoagulation in patients with atrial fibrillation.

Explain the mechanisms underlying AV nodal reentrant tachycardia (AVNRT), AV reentrant tachycardia (AVRT), atrial fibrillation, and ventricular tachycardia.

Skill

Classify tachyarrhythmia by QRS width (narrow vs. wide) and rhythm regularity (regular vs. irregular) to narrow the differential diagnosis.

Select an appropriate rate- or rhythm-control strategy and anticoagulation plan for a patient with atrial fibrillation based on CHA2DS2-VASc score.

Formulate an emergency management plan for a hemodynamically unstable patient with tachyarrhythmia.

Main Subtopics

- Overview and Classification
- Bradyarrhythmia
- Tachyarrhythmia
 - Supraventricular Tachyarrhythmias
 - Atrial Fibrillation
 - Ventricular Tachyarrhythmias
- Wolff-Parkinson-White Syndrome and Pre-excited Atrial Fibrillation (management principles and contraindicated agents)
- Non-Pharmacologic Management (vagal maneuvers, electrical and chemical cardioversion, defibrillation, radiofrequency catheter ablation, pacemaker therapy)

Case scenario :

A 16-year-old boy was brought to the emergency department because of palpitations and lightheadedness that began 16 hours ago. He had been binge drinking the night before. He was sedated and electrically cardioverted. An ECG that was recorded following cardioversion showed a regular rhythm with a heart rate of 80/min, a shortened PR interval, and wide QRS complexes with a slurred upstroke. After regaining consciousness, he was admitted for observation. Serum concentration of creatinine and electrolytes were measured to be within the reference range. Twelve hours after cardioversion, the patient again develops palpitations. He does not have lightheadedness or chest pain. His temperature is 37.1°C (98.8°F), pulse is 220/min, respirations are 20/min, and blood pressure is 112/84 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. Physical examination shows no abnormalities. A newly recorded ECG shows a regular rhythm, a shortened PR interval, and wide, monomorphic QRS complexes.

Case problem :

Which of the following is the most appropriate next step in management?

- A. Perform synchronized electrical cardioversion
- B. Administer magnesium sulfate
- C. Administer verapamil
- D. Administer atenolol
- E. Administer procainamide
- F. Administer adenosine

Cardiac Arrhythmias

Accelerated, slowed, or irregular heart rates caused by abnormalities in the electrical impulses of the myocardium.

Bradyarrhythmia

Definition $< 60/\text{minutes}$.

Tachyarrhythmia

Definition $> 100/\text{minute}$)

Causes and mechanisms of Brady-dysrhythmia

Physiological:

- particularly in athletes

Cardiac:

- Sinus node dysfunction (sick sinus syndrome)
- Underlying cardiovascular disease : IHD, DHD
- Increased vagal tone
- Sinoatrial and AV conduction disease
- Right coronary infarction

Extra cardiac:

- Drugs: digoxin, beta blockers, calcium channel blockers

Causes and mechanisms of Tachy-dysrhythmia

Physiological response in healthy individuals

Cardiac:

- Underlying **cardiovascular disease**
(VHD - Myocarditis - Myocardial infarction- CHF)
- Sympathetic activation or vagal withdrawal on the SA node
- Discharge from a single or multiple ectopic foci in the atrium
- Accessory pathway and macroreentrant rhythms

Inherited arrhythmia syndromes (**channelopathies**) :

- ✓ Congenital long QT syndrome
- ✓ Brugada syndrome
- ✓ catecholaminergic polymorphic ventricular tachycardia (CPVT)

Extra cardiac:

- Electrolyte imbalances
- Associated disorders (COPD exacerbation, PE, Thyrotoxicosis)
- Drugs : **Digitalis toxicity** – theophylline

Clinical Picture of Tachy-dysrhythmia

- **Asymptomatic.**
- **When symptoms do occur, they include**
 - Palpitations
 - Light-headedness
 - Shortness of breath
 - Syncope

Classification

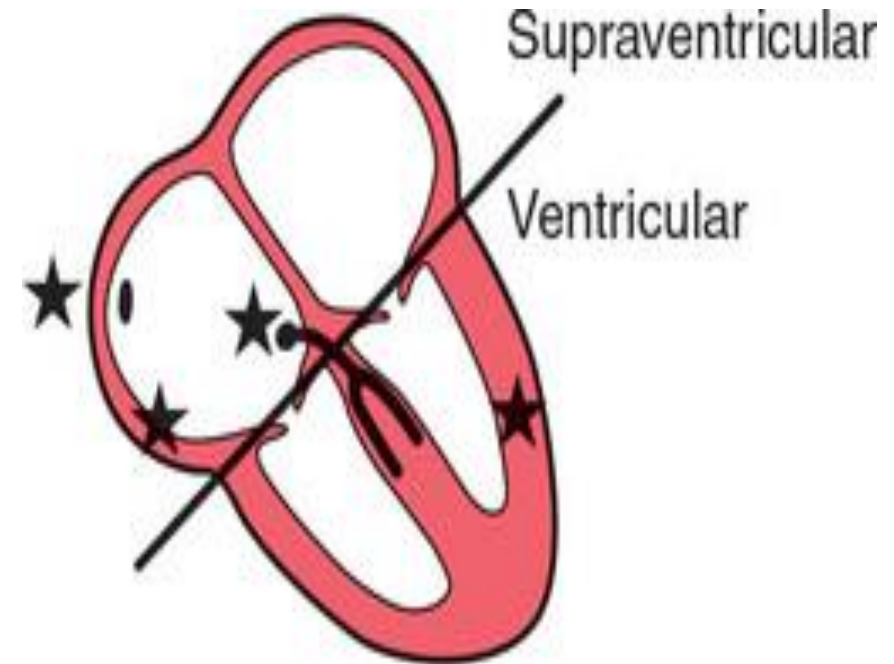
Supraventricular arrhythmias

Definition: arrhythmias that originate in the

- ✓ sinoatrial node,
- ✓ atrial myocardium, or
- ✓ atrioventricular node

Ventricular arrhythmias

Definition: arrhythmias that originate below the atrioventricular node (**wide QRS complex**)



Types of bradyarrhythmia

Atrial origin

- Respiratory sinus arrhythmia
- Sinus bradycardia
- Sinus pause or arrest
- Tachycardia-bradycardia syndrome

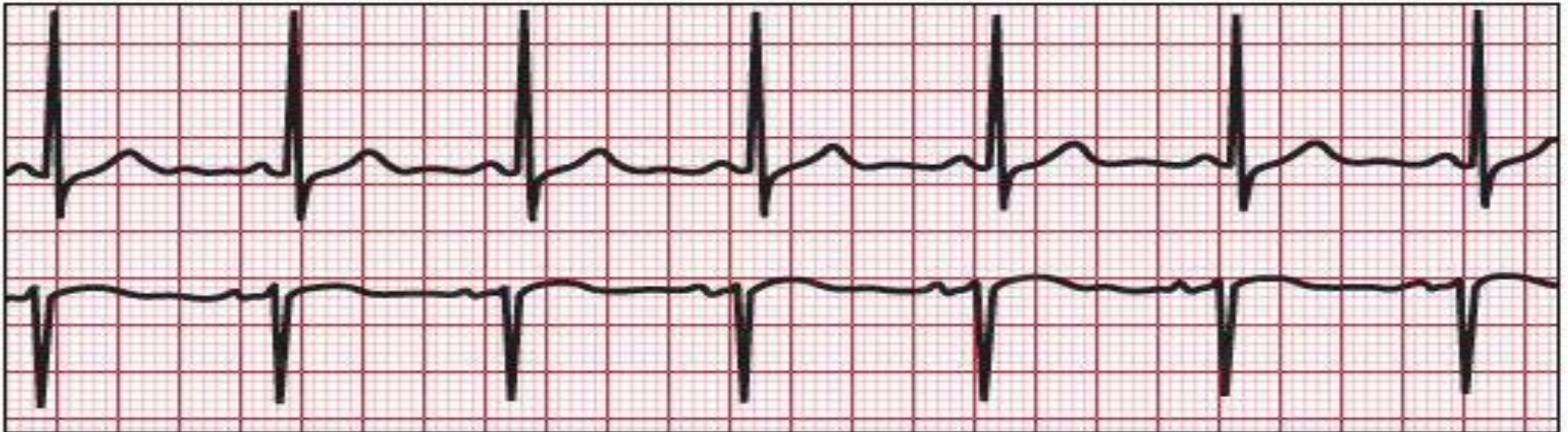
AV node origin

Atrioventricular block

- First-degree block
- Second-degree block
 - Mobitz type I (Wenckebach)
 - Mobitz type II
- Third-degree block

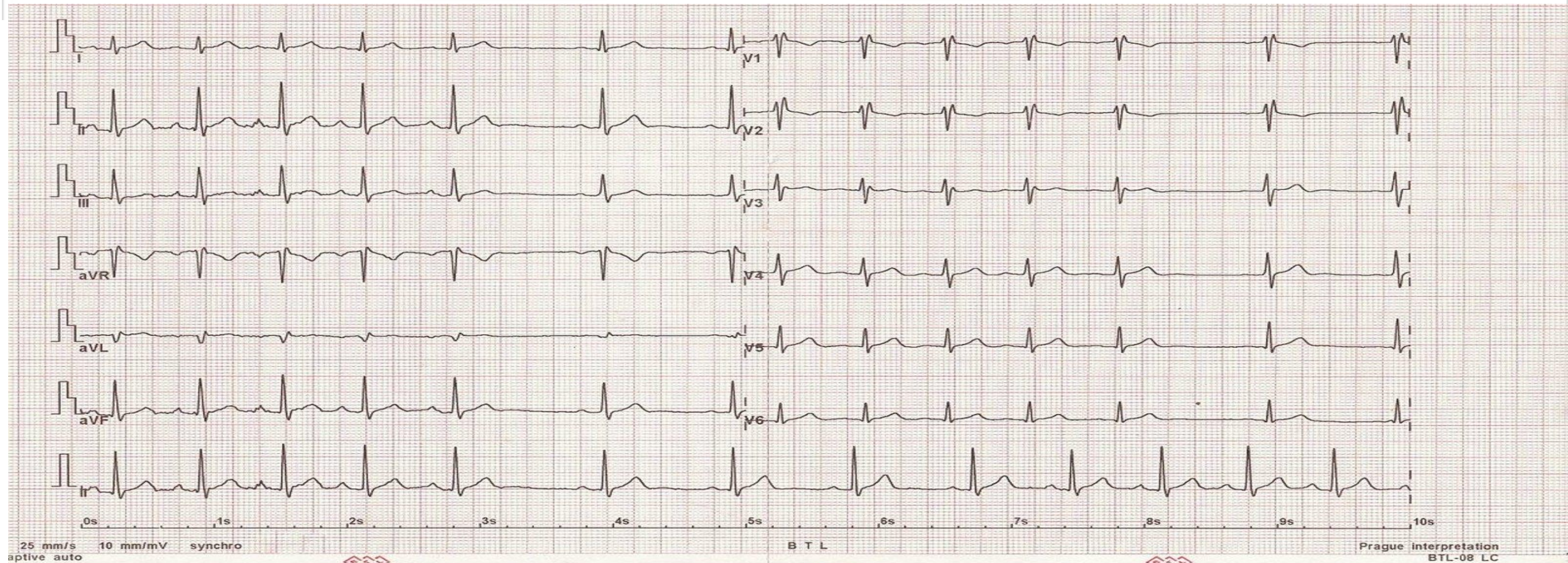
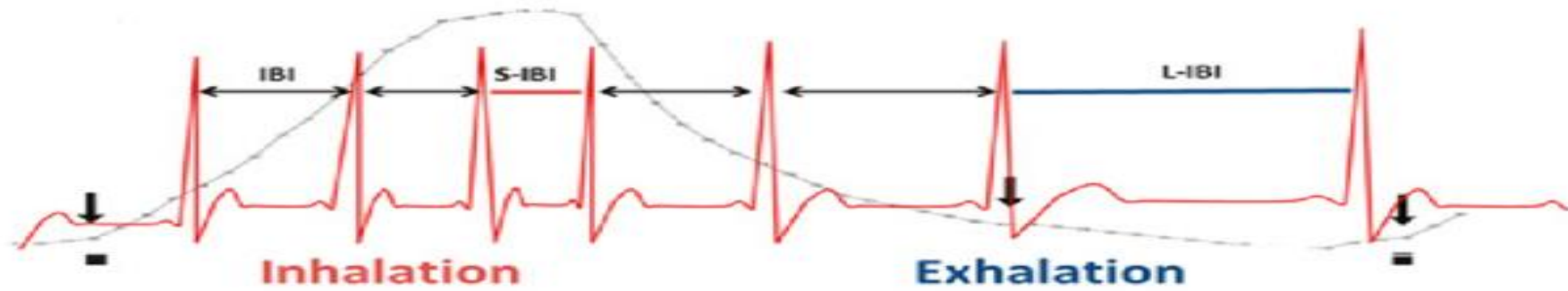
Normal Rhythm

Normal Sinus Rhythm



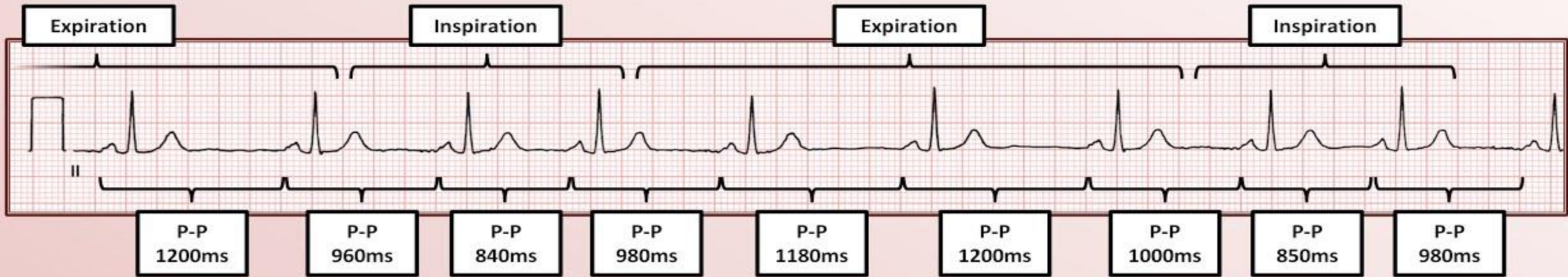
Heart Rate	Rhythm	P Wave	PR interval (in seconds)	QRS (in seconds)
60-100 bpm	Regular	Before each QRS, identical	.12 to .20	<.12

Respiratory sinus arrhythmia



Respiratory sinus arrhythmia

Respiratory Sinus Arrhythmia



Non-Respiratory Sinus Arrhythmia

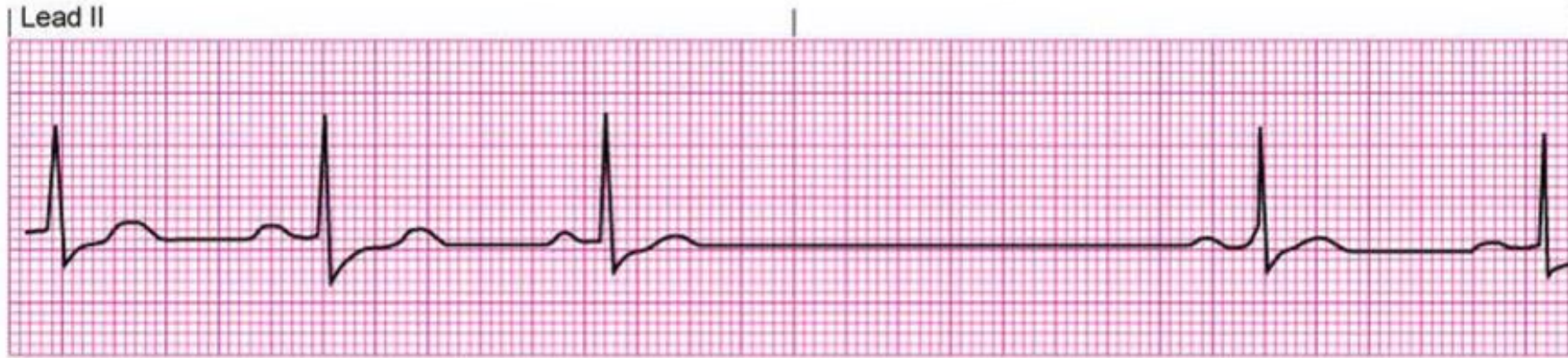


Unrelated to respiration, and without a discernable pattern

Sinus bradycardia



Sinus pause or arrest



A Sinus arrest

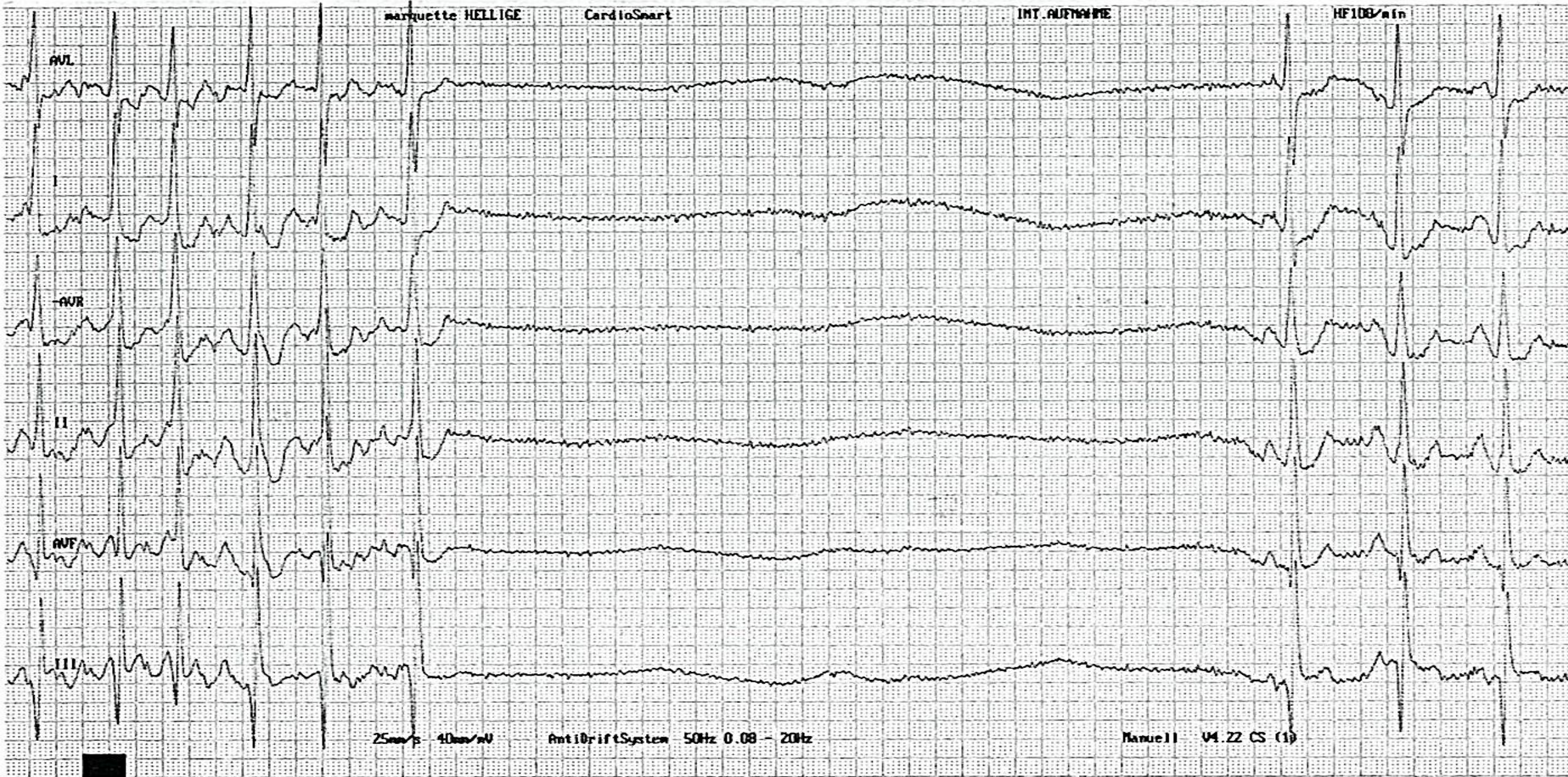


B Sinus exit block

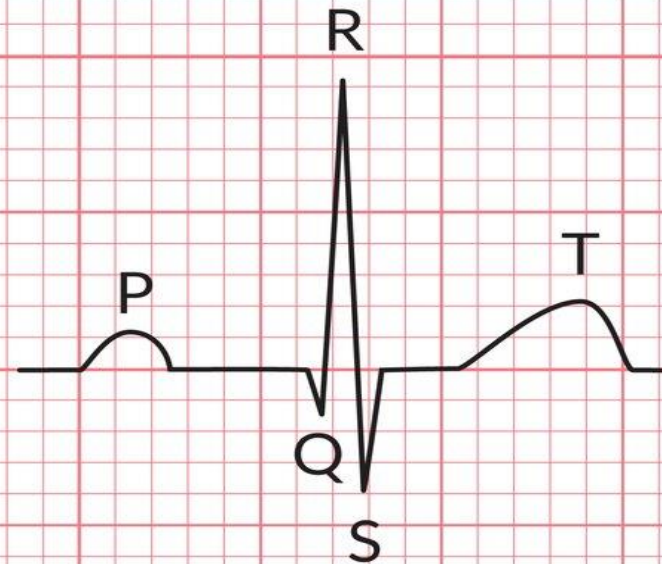
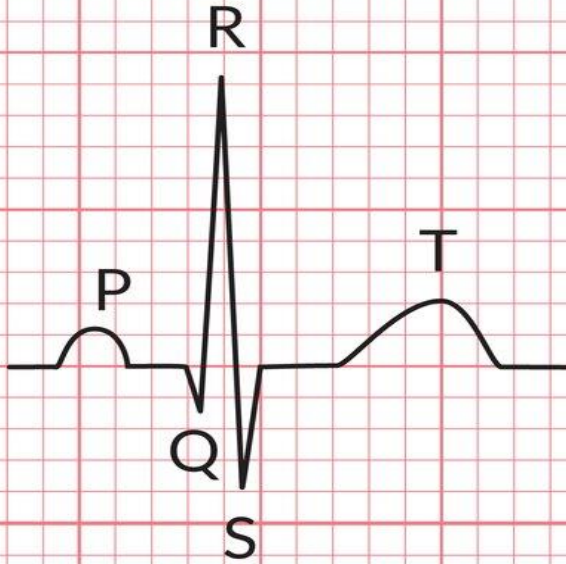
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Figure 7-28. A, Sinus arrest. B, Sinus exit block.

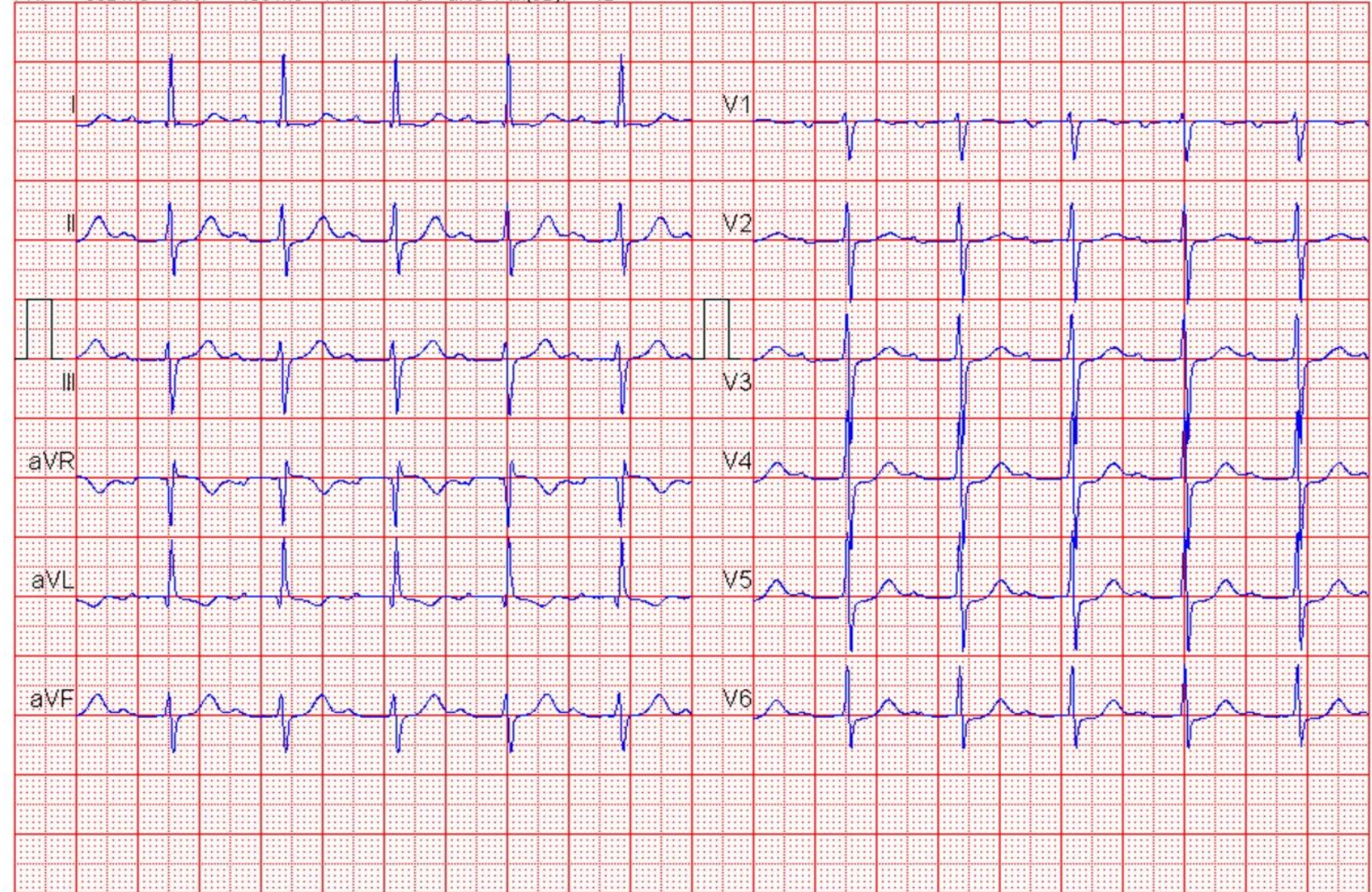
Tachycardia-bradycardia syndrome



First-degree block



HR: 81 QRS: 106 ms P ax: 69
Pd: 120 ms QT: 400 ms QRS ax: -22
PR: 302 ms QTc: 465 ms T ax: 79 QRS-T ax(3D): 42

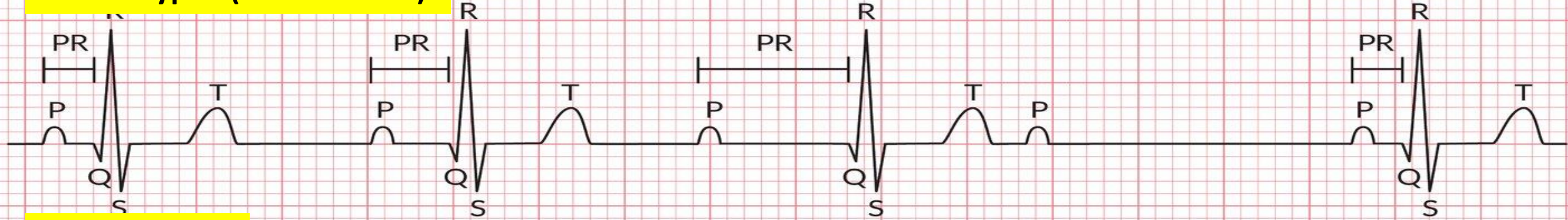


T: 7:23 - 7:27

25 mm/s 10 mm/mV

Second-degree block

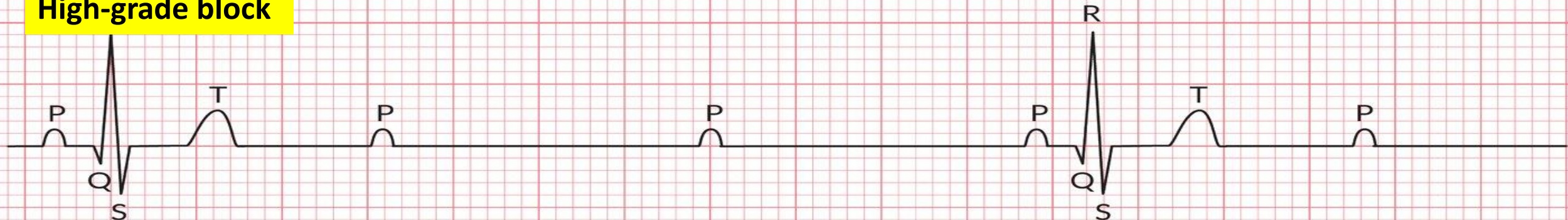
Mobitz type I (Wenckebach)



Mobitz type II

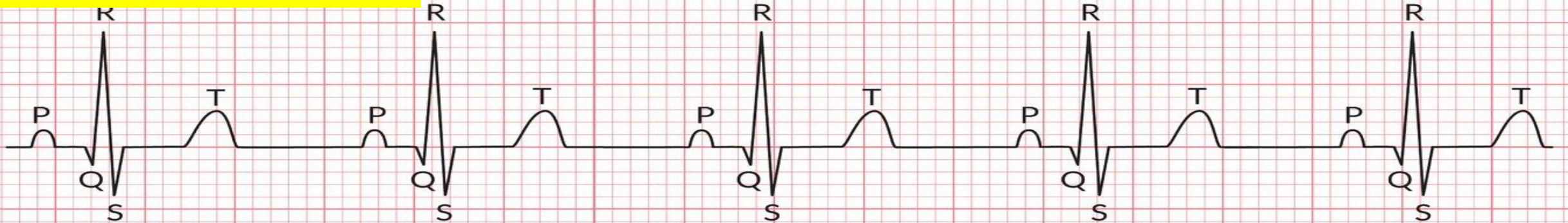


High-grade block



Third-degree atrioventricular block (complete heart block)

Sinus rhythm: rate ~ 83/min



Third-degree AV block with junctional escape rhythm: rate ~ 45/min



Third-degree AV block with ventricular escape rhythm : rate ~ 45/min



Management of bradycardia

Assess appropriateness for clinical condition.
Heart rate typically $< 50/\text{min}$ if bradyarrhythmia.

Identify and treat underlying cause

- Maintain patent airway; assist breathing as necessary*
- If the patient is short of breath, administer oxygen
- Oxygen as indicated
- Cardiac monitor to identify rhythm; monitor blood pressure and oximetry
- IV access
- 12-lead ECG if available; don't delay therapy

Persistent bradyarrhythmia causing:

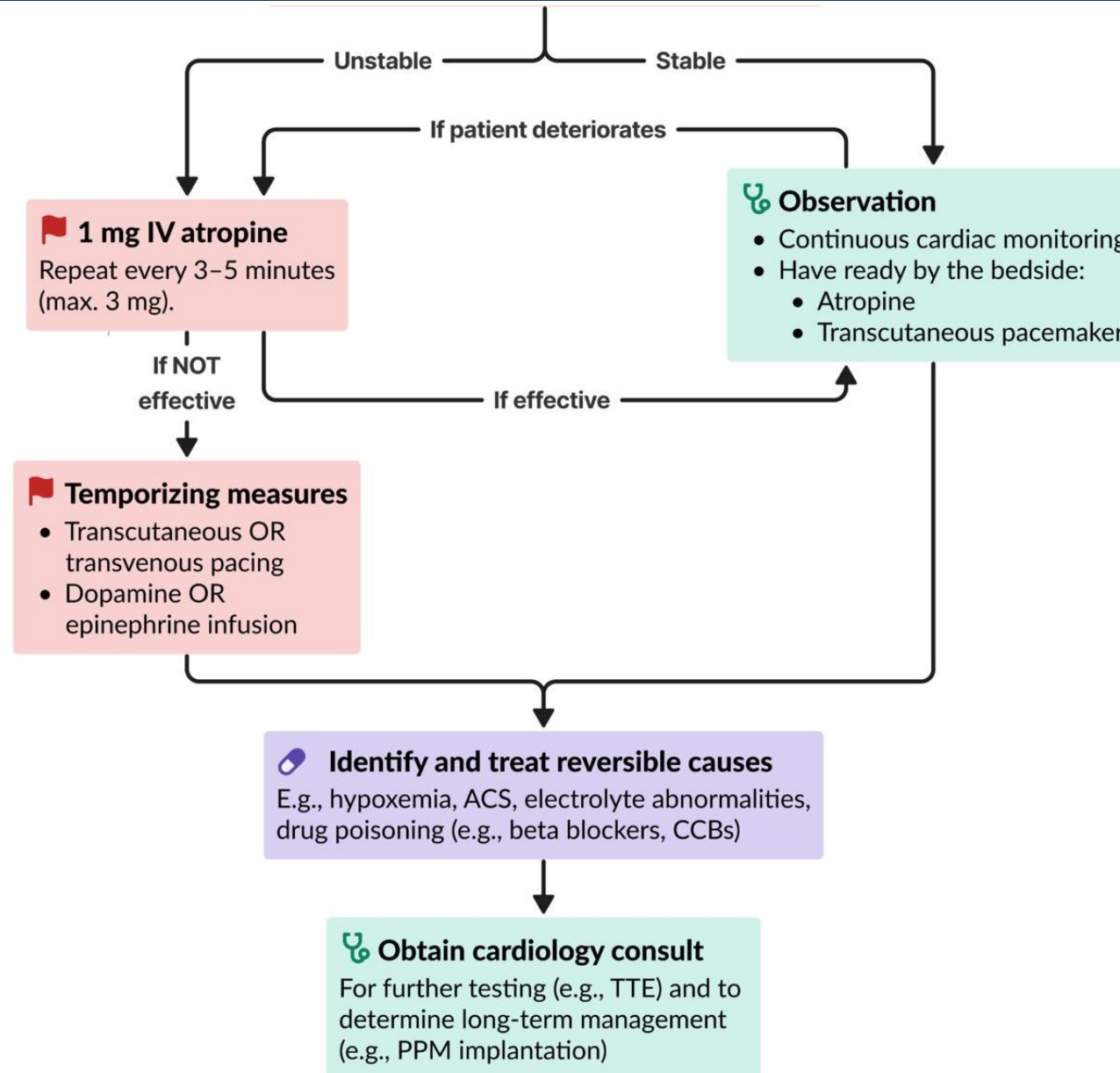
- Hypotension?
- Acutely altered mental status?
- Signs of shock?
- Ischemic chest discomfort?
- Acute heart failure?

Monitor
and
observe

N

Y

Management of Bradycardia



Types of Tachyarrhythmia

Supraventricular arrhythmias

Atrial origin

- Supraventricular premature beats
- Sinus tachycardia
- Atrial flutter
- Atrial fibrillation
- Atrial tachycardia (~ 5%)
- Multifocal atrial tachycardia (MAT)

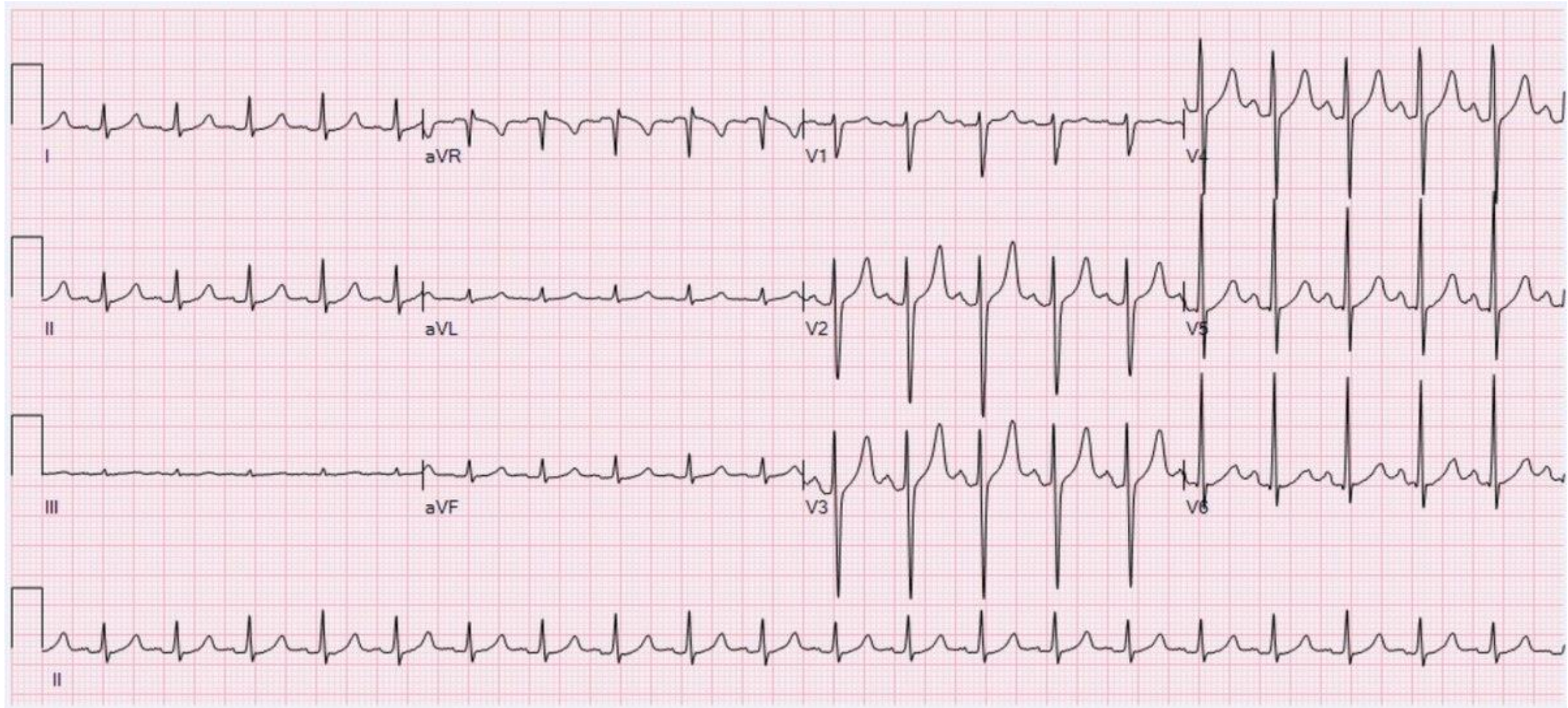
AV node origin

- Atrioventricular reentry tachycardia (AVRT)
 - Orthodromic AVRT: narrow QRS complex
 - Antidromic AVRT: wide QRS complex with delta waves
- AV nodal reentry tachycardia (AVNRT)
- Junctional tachycardia

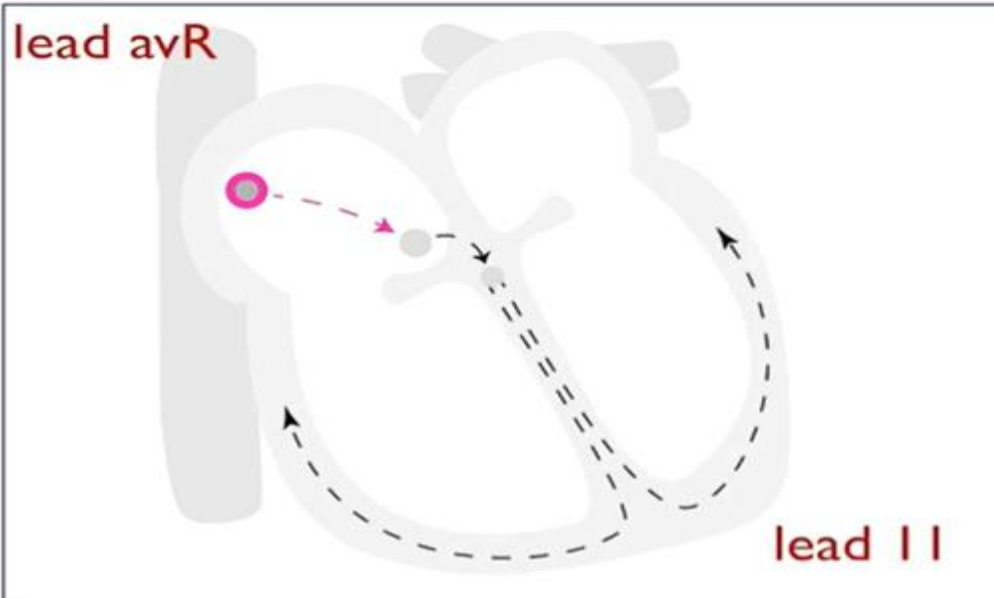
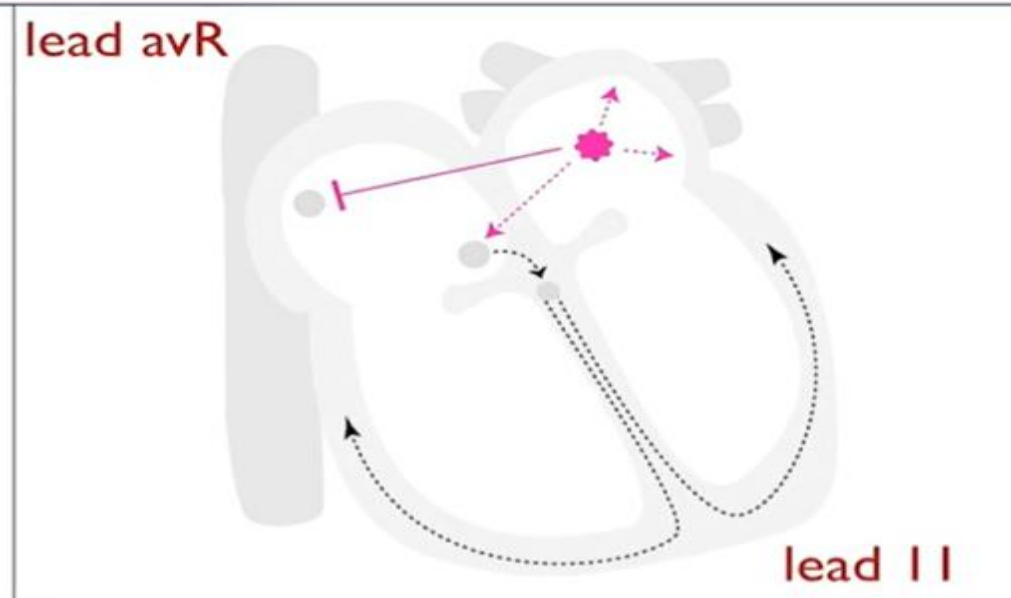
Ventricular tachycardia

- Monomorphic VT (most common): single QRS morphology
- Polymorphic VT: multiple QRS morphologies
- Torsade de pointes tachycardia
- Ventricular fibrillation

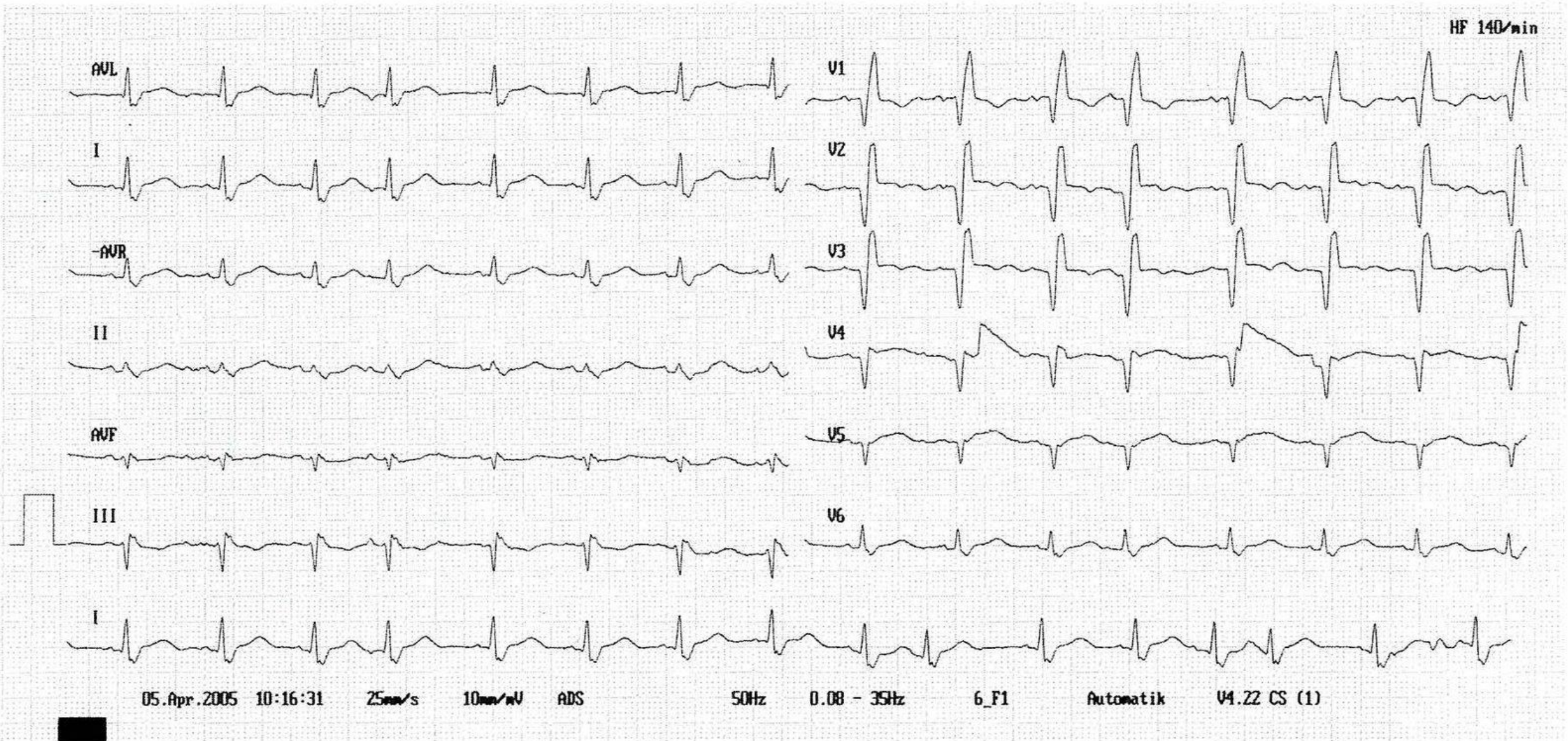
Sinus tachycardia



Sinus tachycardia VS Focal atrial tachycardia

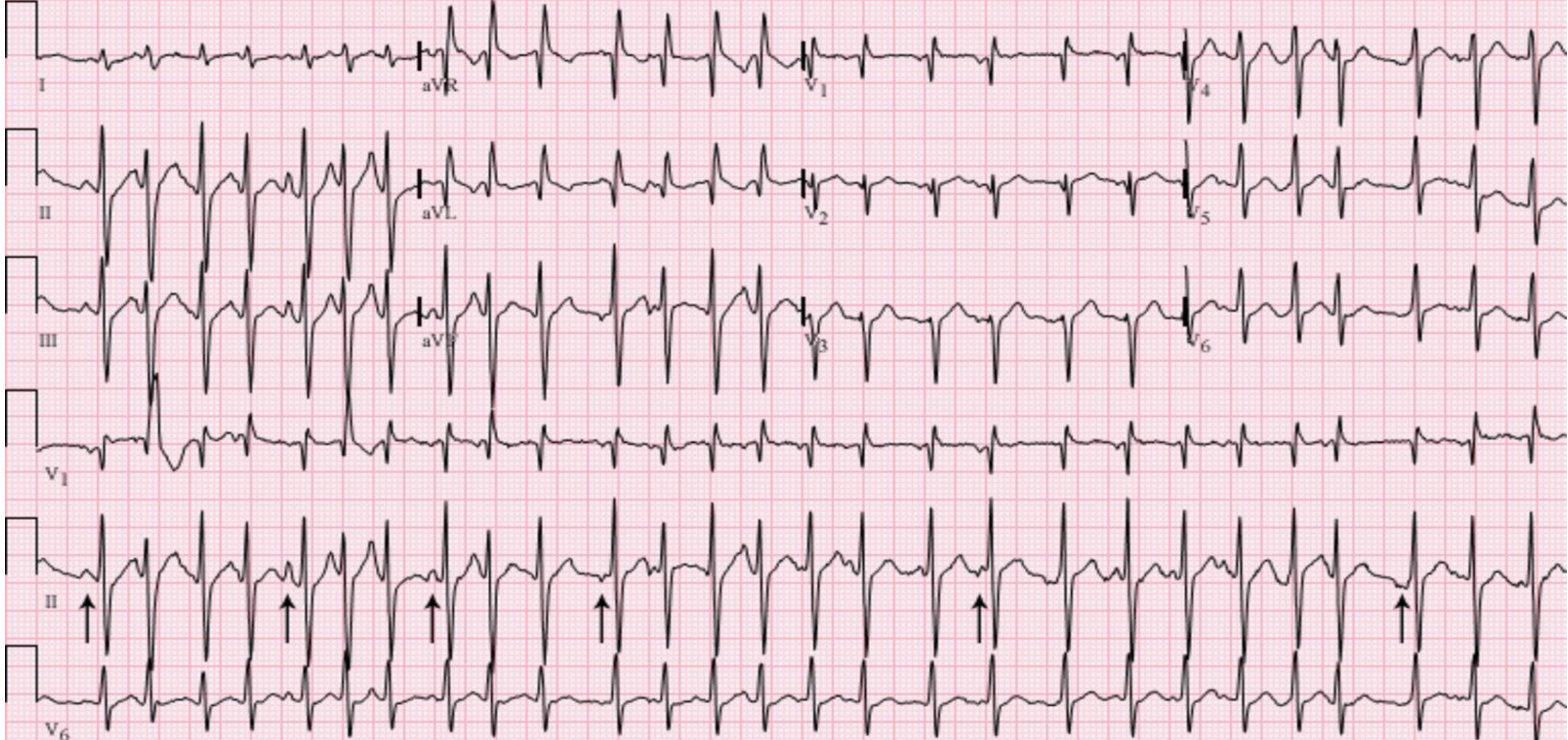
	
Sinus Pacemaker Sinus P wave 1- Round in shape. 2- Normal axis upright in lead II 3- Identical before each QRS 4- Atrial Rate (P- P) not exceed 150bpm	Atrial Pacemaker Atrial P wave 1- not round in shape. 2- Abnormal axis inverted in lead II 3- Identical before each QRS 4- Atrial Rate (P- P) exceed 150bpm

Supraventricular premature beats



Multifocal atrial tachycardia (MAT)

Courtesy of Jason E. Roediger, CCT, CRAT

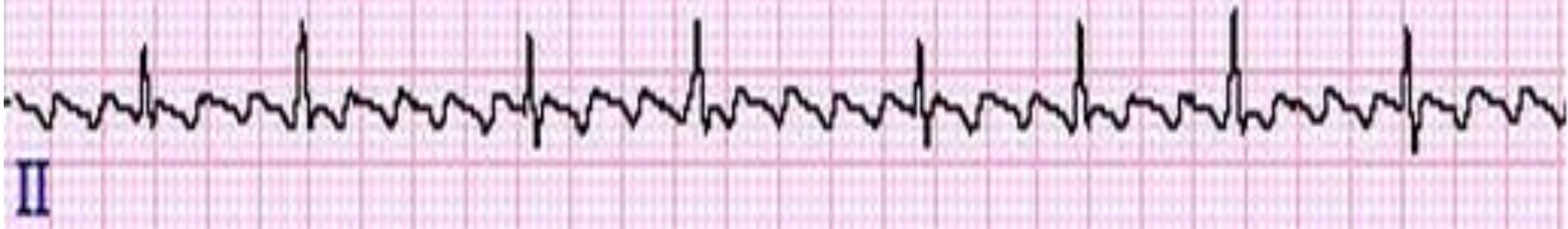


Atrial flutter and Atrial fibrillation

Atrial Fibrillation - fibrillatory waves



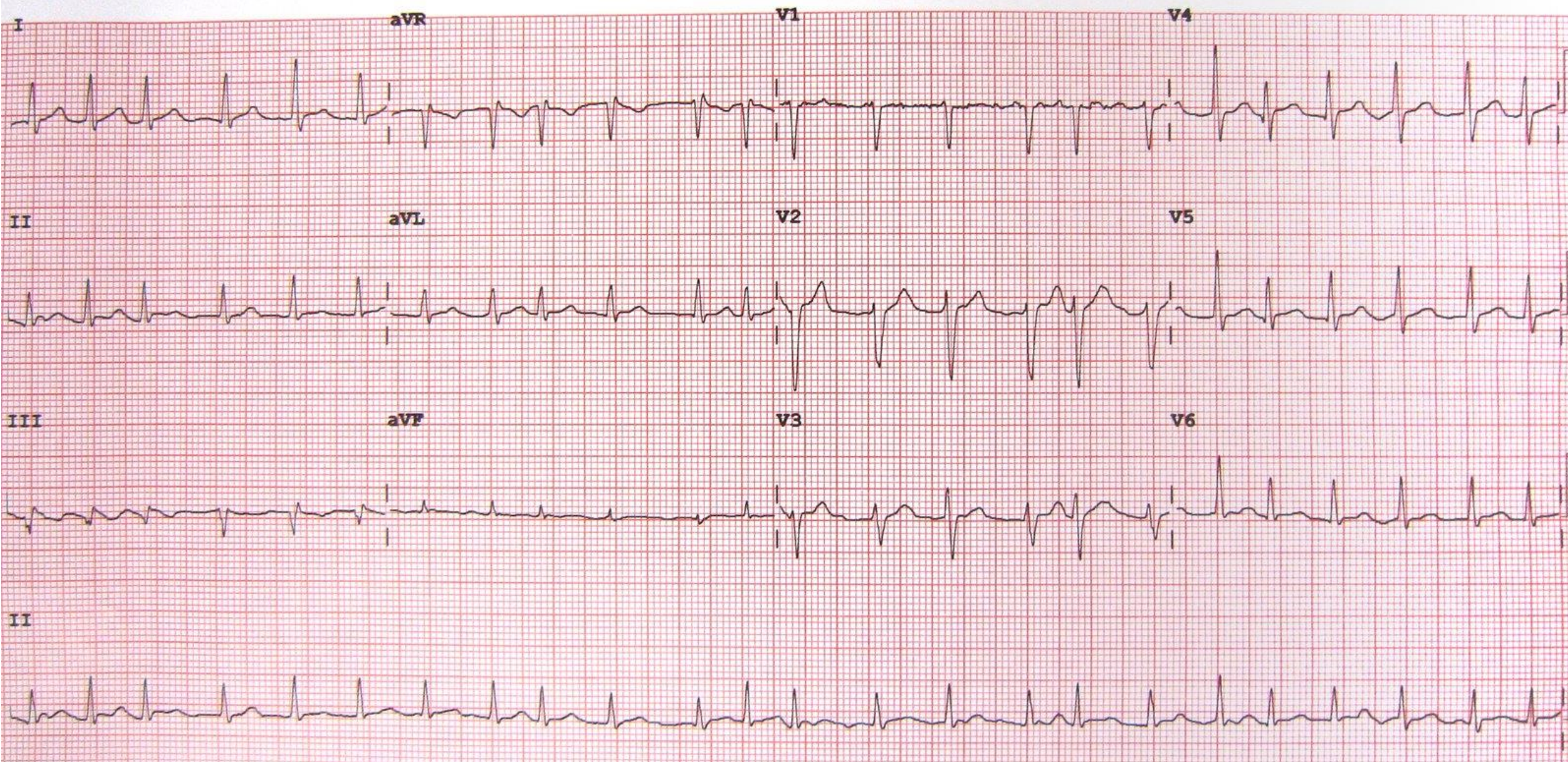
Atrial Flutter - sawtooth pattern



Atrial fibrillation

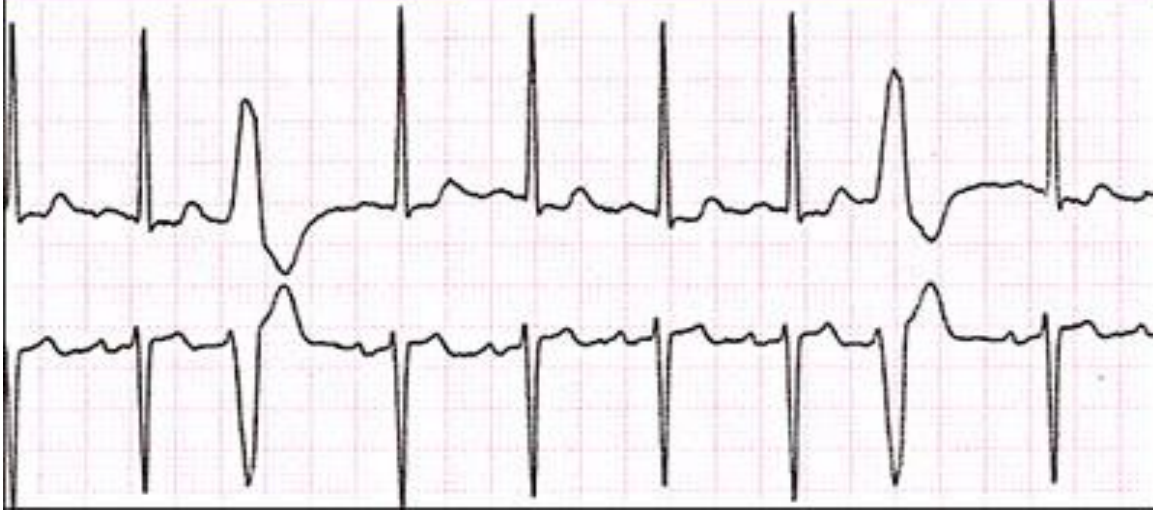


Atrial fibrillation with rapid ventricular response



PVCs and ventricular tachycardia

Unifocal PVC's: identical shapes
Note: A single PVC is labeled isolated



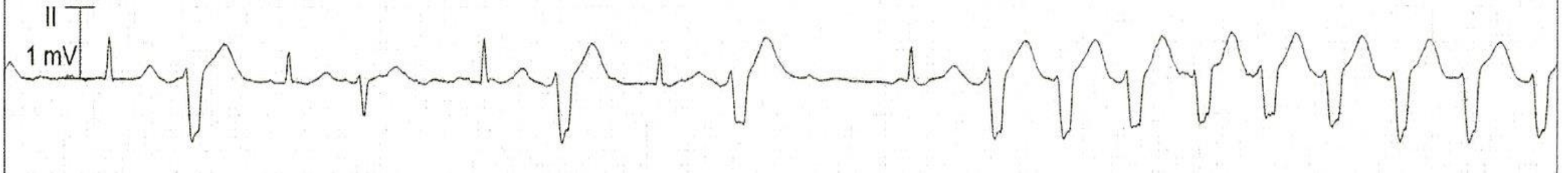
Multifocal PVC's: more than one shape



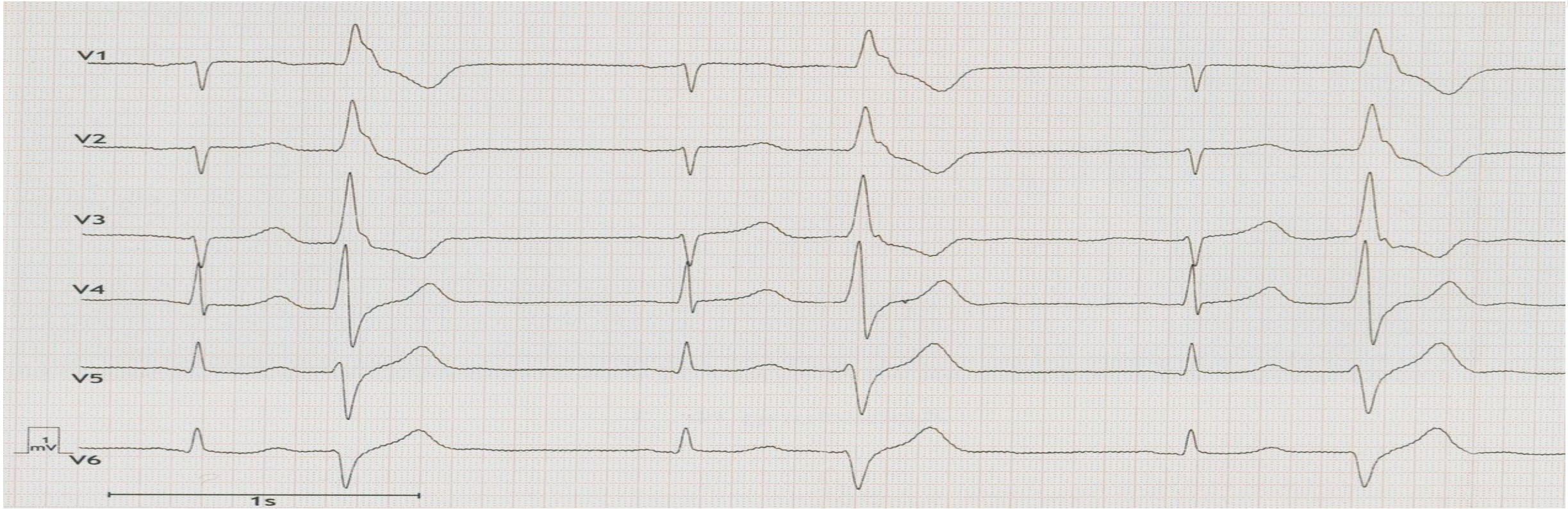
Kennummer:

Bett3

HF 93 %SpO2 96 VES 9 NBP 100/ 44(65) PULS ? RESP 23



Ventricular bigeminy



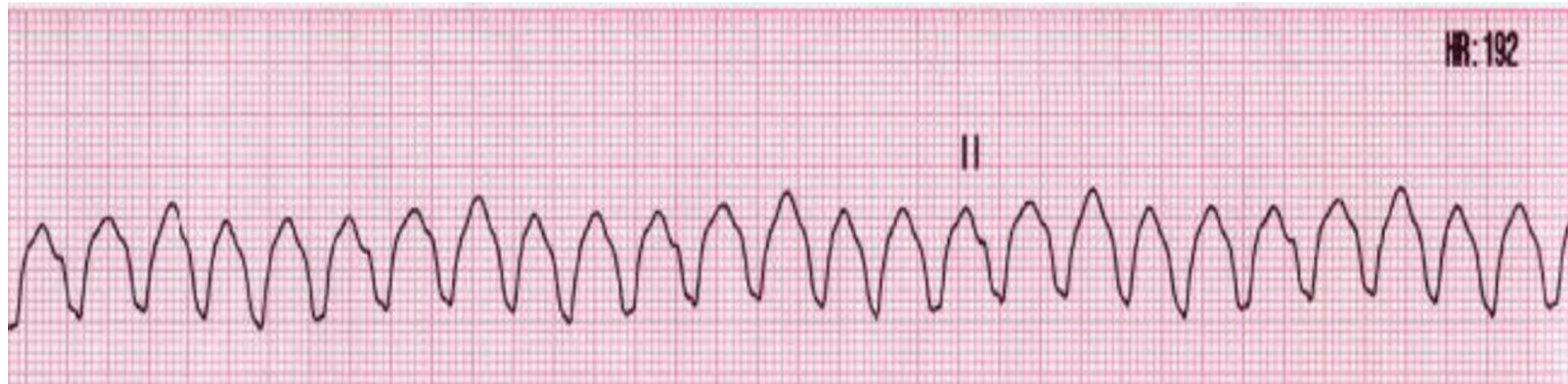
Treatment

- Most patients (asymptomatic) do not require any treatment
- Treat any underlying disease
- Only treat frequent and significantly symptomatic PVCs
- Antiarrhythmic therapy(BB)

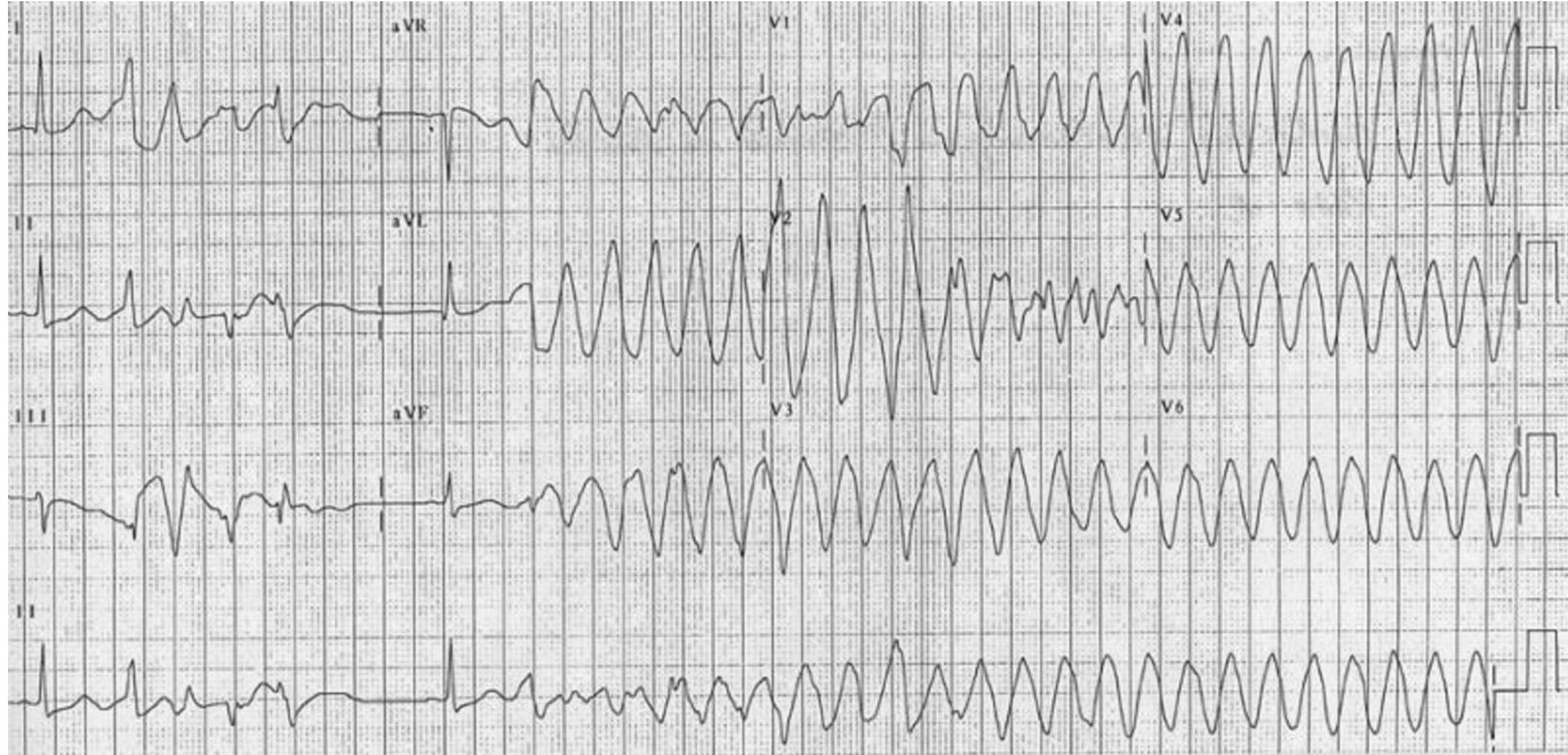
Frequent premature ventricular contractions



Monomorphic ventricular tachycardia

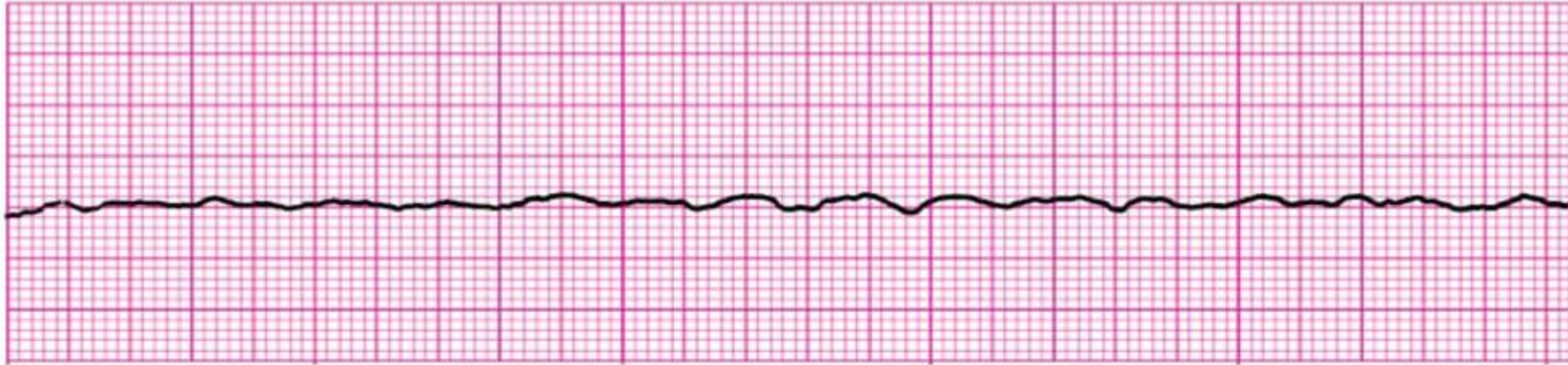


Torsades de pointes (TdP) polymorphic ventricular tachycardia

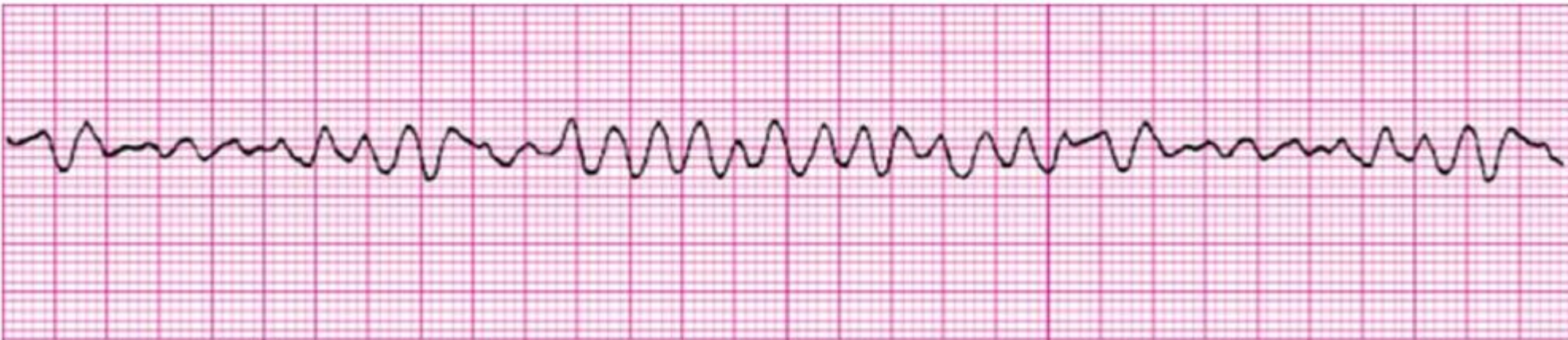


Ventricular fibrillation

Fine VF Low amplitude waves (less than 3 mm)



Coarse VF Waves more easily visible (greater than 3 mm)

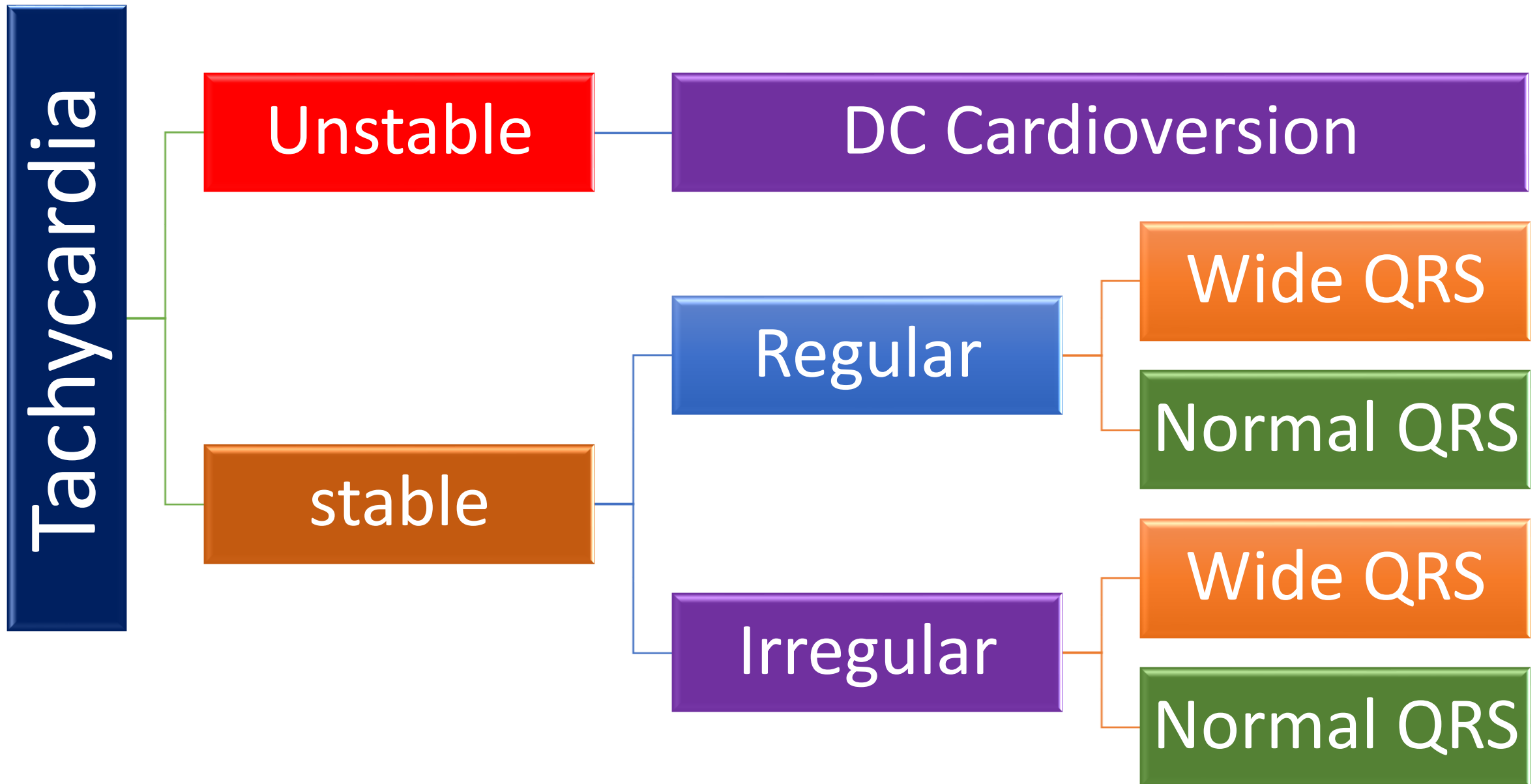


Ventricular fibrillation

Courtesy of Jason E. Roediger, CCT, CRAT



Management of tachycardia



Signs of hemodynamic instability

- Chest pain
- Altered mental status
- Acute pulmonary edema
- Hypotension
- Cardiogenic shock

Four groups

Narrow regular

Narrow irregular

Wide regular

Wide irregular

Types of Tachyarrhythmia

Tachyarrhythmia

Regular

Narrow QRS:

ST, SVT, A Flutter, A tachycardia

Wide QRS:

VT , (+ AP or BBB)

Irregular

Narrow QRS:

AF, MAT

Wide QRS:

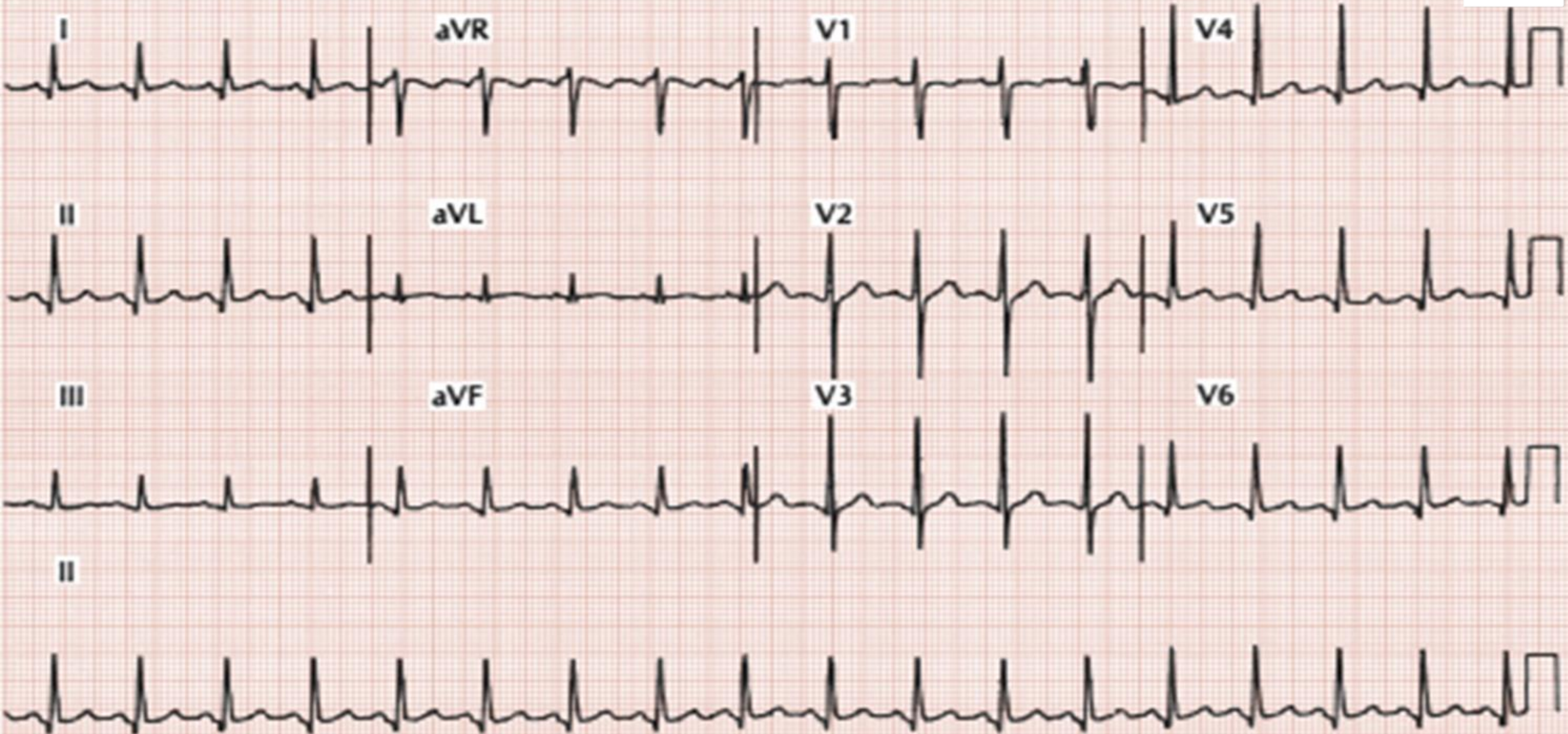
AF+(AP or BBB) , VF

Narrow regular

Causes:

- Sinus tachycardia,
- AVNRT,
- AVRT,
- A.flutter
- A.tachycardia.

Sinus tachycardia



Treatment of sinus tachycardia

- **Asymptomatic Patient**

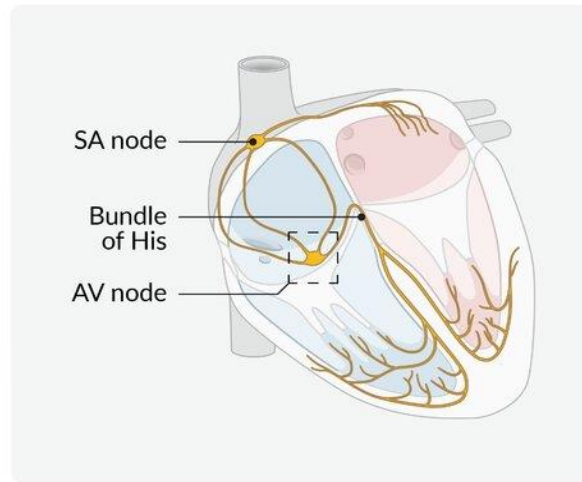
Treatment is not required in asymptomatic individuals without underlying structural heart defects

- **Symptomatic patients**

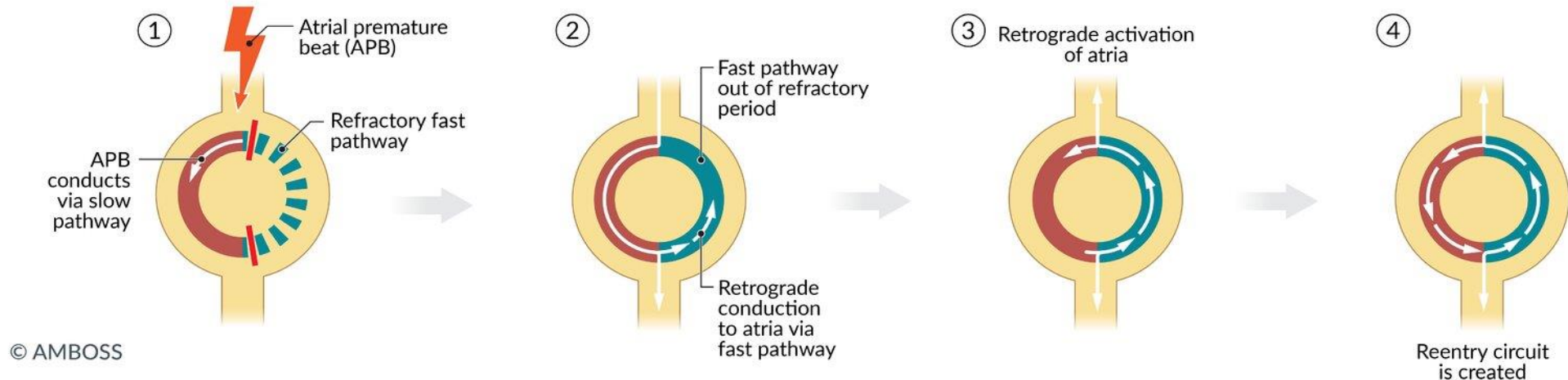
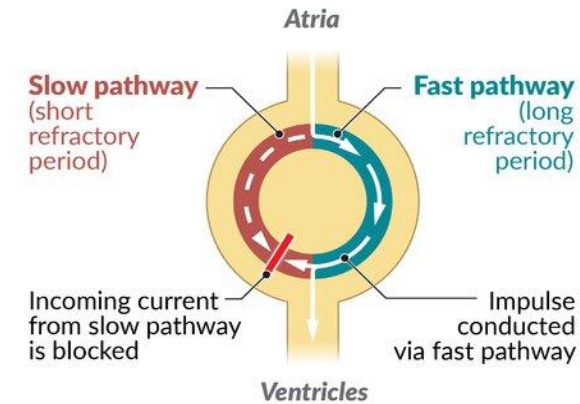
- ✓ Reduce potential triggers like caffeine, alcohol, stress and smoking
- ✓ Treatment of correctable causes like anaemia, thyrotoxicosis and fever
- ✓ Beta blockers in patients with persistent symptoms

Management of SVT (AVRT and AVNRT)

AVRT and AVNRT



Dual AV nodal physiology



Management of SVT (AVRT and AVNRT)

AVRT and AVNRT

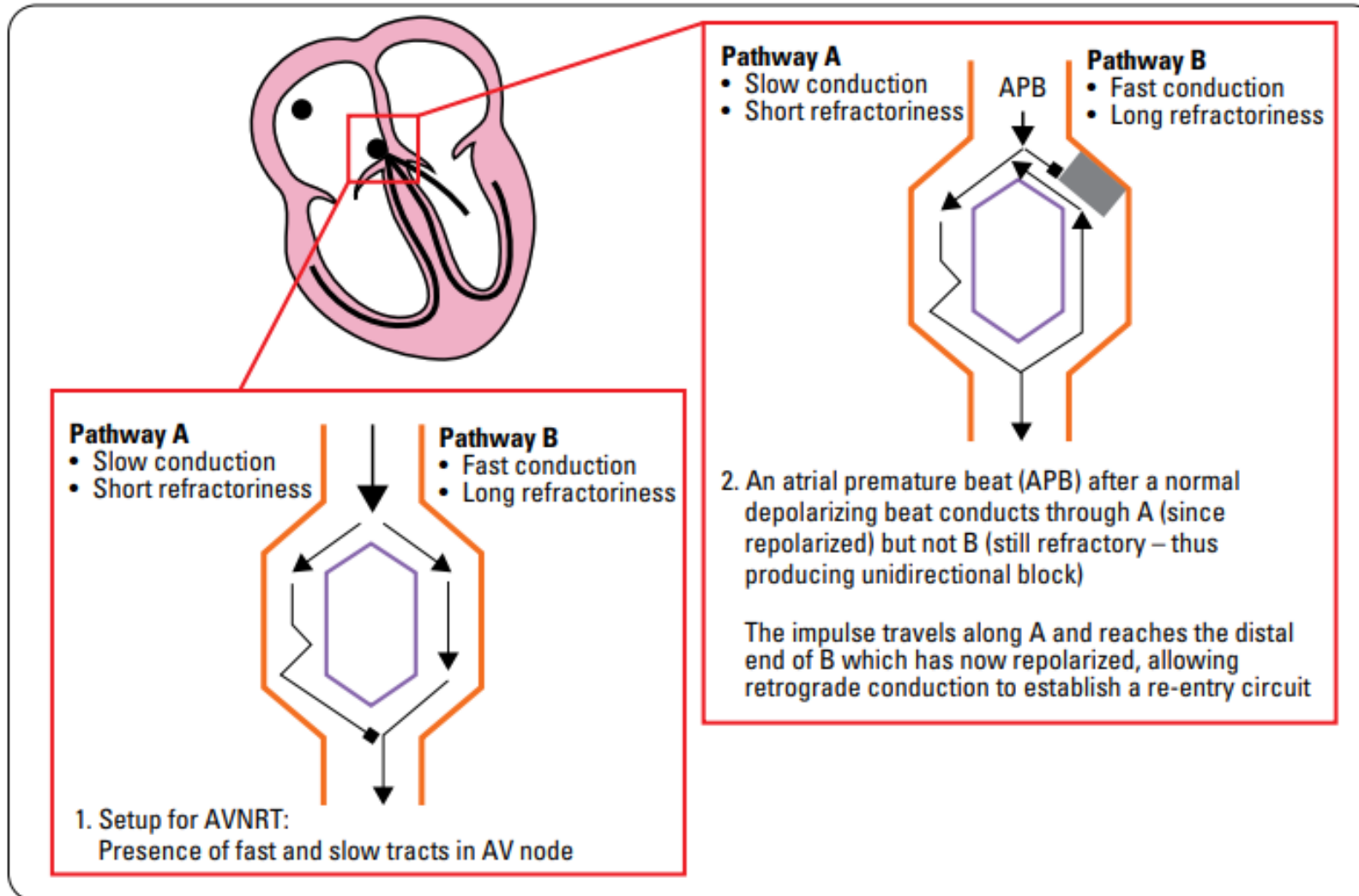


Figure 28. Mechanism for AVNRT

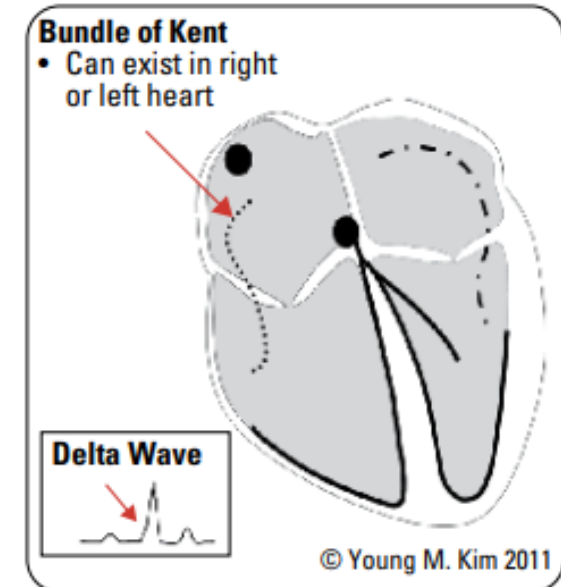


Figure 29. Accessory pathway conduction in WPW. Early ventricular activation leads to the appearance of a delta wave (slurred upstroke of the QRS) on the ECG before usual conduction across the AV node

Management of SVT (AVRT and AVNRT)

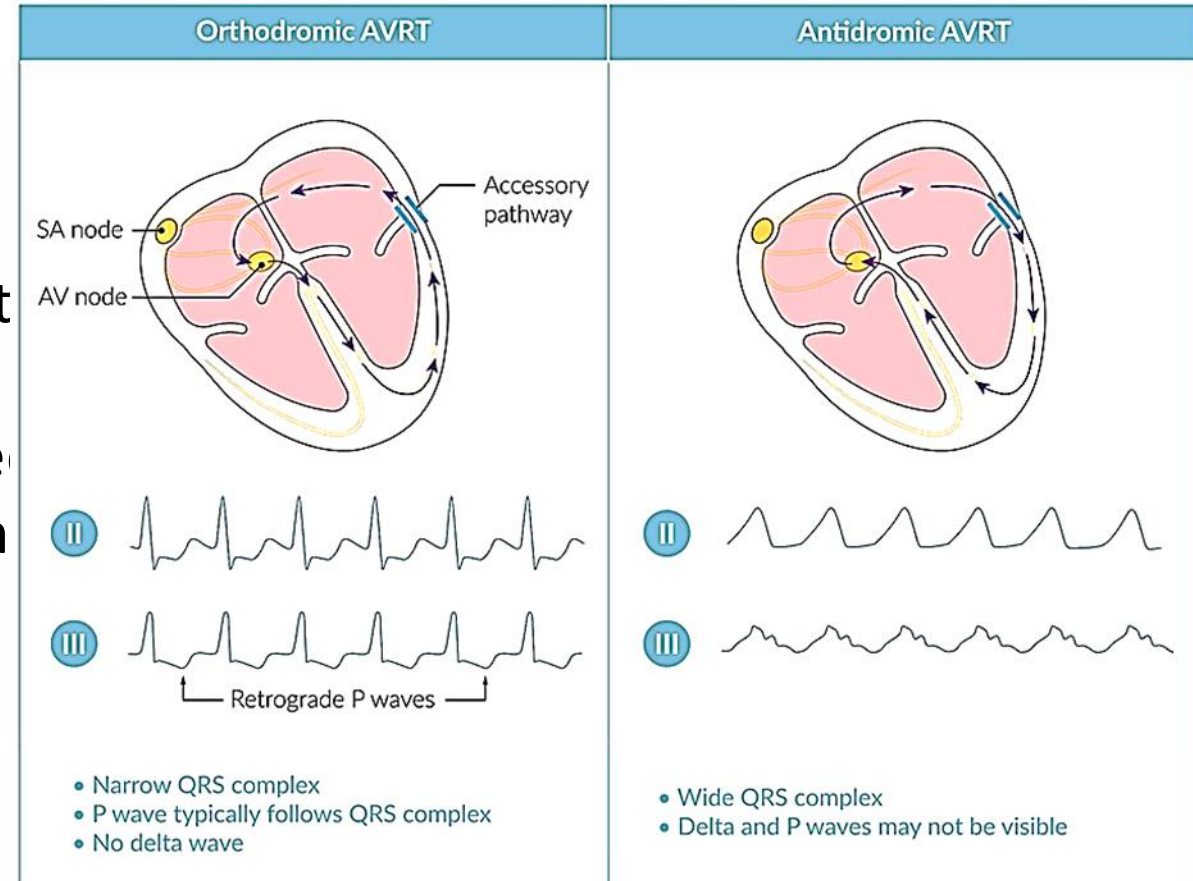
AV Nodal Re-Entrant Tachycardia

- Re-entrant circuit using dual pathways (fast conducting β -fbres and slow conducting α -fbres) within or near the AV node; often found in the absence of structural heart disease
 - cause is commonly idiopathic, although familial AVNRT has been reported
- Sudden onset and offset with patients often describing “neck pounding”.
- Fast regular rhythm of 150-250 bpm
- Usually initiated by a supraventricular or ventricular premature beat
- AVNRT accounts for 60-70% of all paroxysmal SVTs
- Retrograde P waves may be seen but are usually lost in the QRS complex

Management of SVT (AVRT and AVNRT)

AV Re-Entrant Tachycardia (AVRT)

- Re-entrant loop in antegrade via normal conduction system and retrograde via accessory pathway
- Usually in patients with an antegradely conducting bypass tract (WPW); may also occur if there is an exclusively retrogradely conducting (i.e. concealed) bypass tract - in these cases the ECG is normal and there are no delta waves
- Initiated by a premature atrial or ventricular complex



Management of SVT (AVRT and AVNRT)

AV Nodal Re-Entrant Tachycardia

• Treatment

■ Acute:

- **Electrical cardioversion if patient hemodynamically unstable**
- Valsalva maneuver or carotid sinus pressure technique, (**with precautions ??**)
- Adenosine is first choice if unresponsive to vagal maneuvers; Adenosine 6 mg (**rapid iv - incremental doses**)
- - Isoptin 5mg iv over 3-5 min. .. **Repeated**
- If no response, try metoprolol, digoxin, diltiazem,

■ Long-term:

- **1st line** radiofrequency ablation (>98% cure rate and << 1% complication rate), β -blocker, diltiazem, and digoxin.
- **2nd line** flecainide, propafenone

Management of SVT (AVRT and AVNRT)

AV Re-Entrant Tachycardia (AVRT)

- **Treatment**

- acute: treatment is similar to AVNRT but **avoid long-acting AV nodal blockers** (e.g. digoxin and verapamil)
- long-term: for recurrent arrhythmias, ablation of the bypass tract is recommended
- drugs such as flecainide and procainamide can be used

Paroxysmal supraventricular tachycardia (AVNRT)

Remember: to identify Any Rhythm you must answer 3 Questions???

- What's the Pacemaker??
- What's the Regularity??
- What's the Heart Rate??



Junctional Pacemaker : Absent P wave

Or Retrograde P wave Inverted in lead II and upright in avR
Just before or After QRS complex



Pacemaker

Regularity

Heart rate

QRS

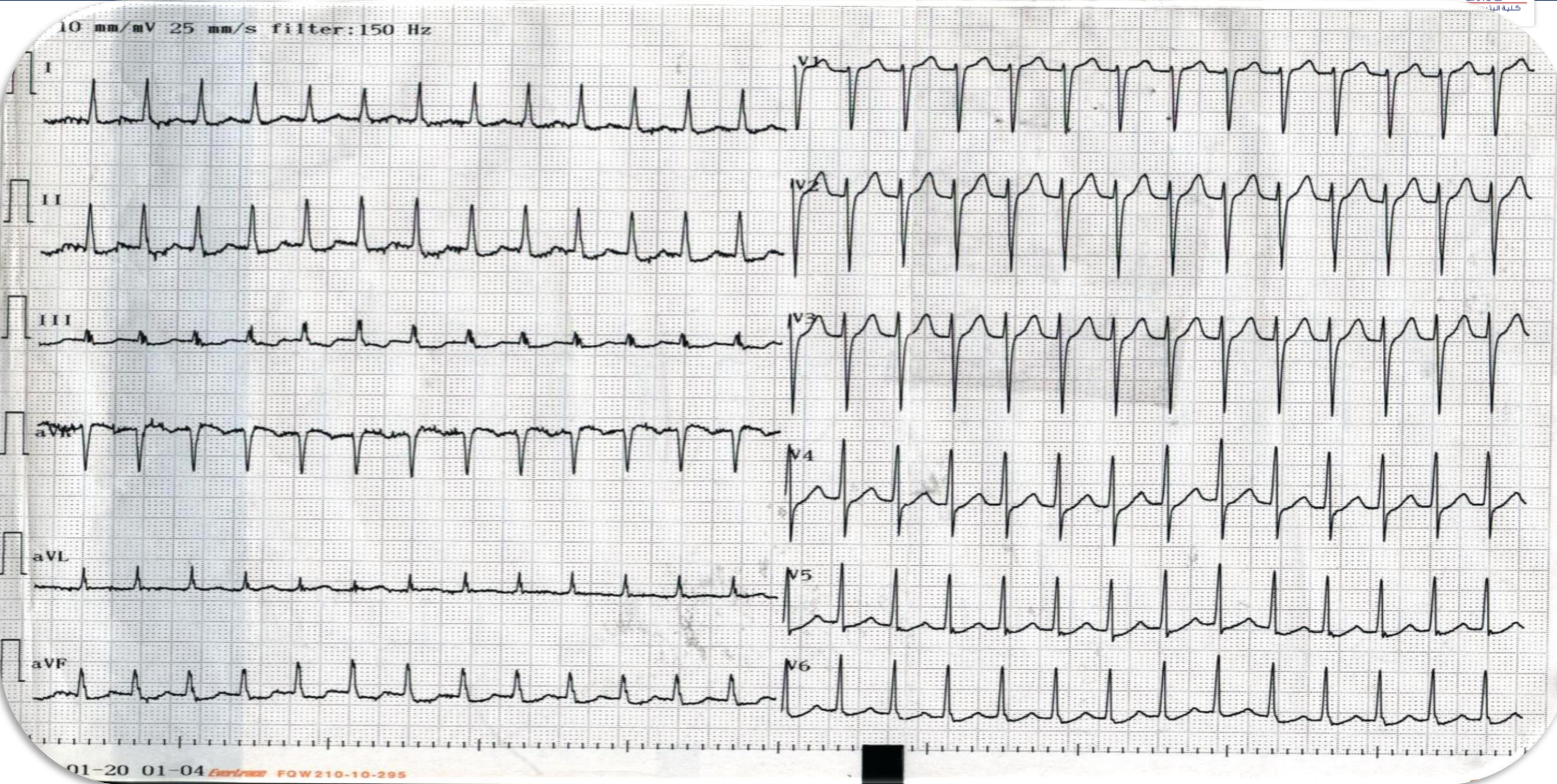
Junctional Pacemaker : Absent P wave
Or Retrograde P wave Inverted in
lead II and upright in avR
Usually seen After QRS complex

Regular

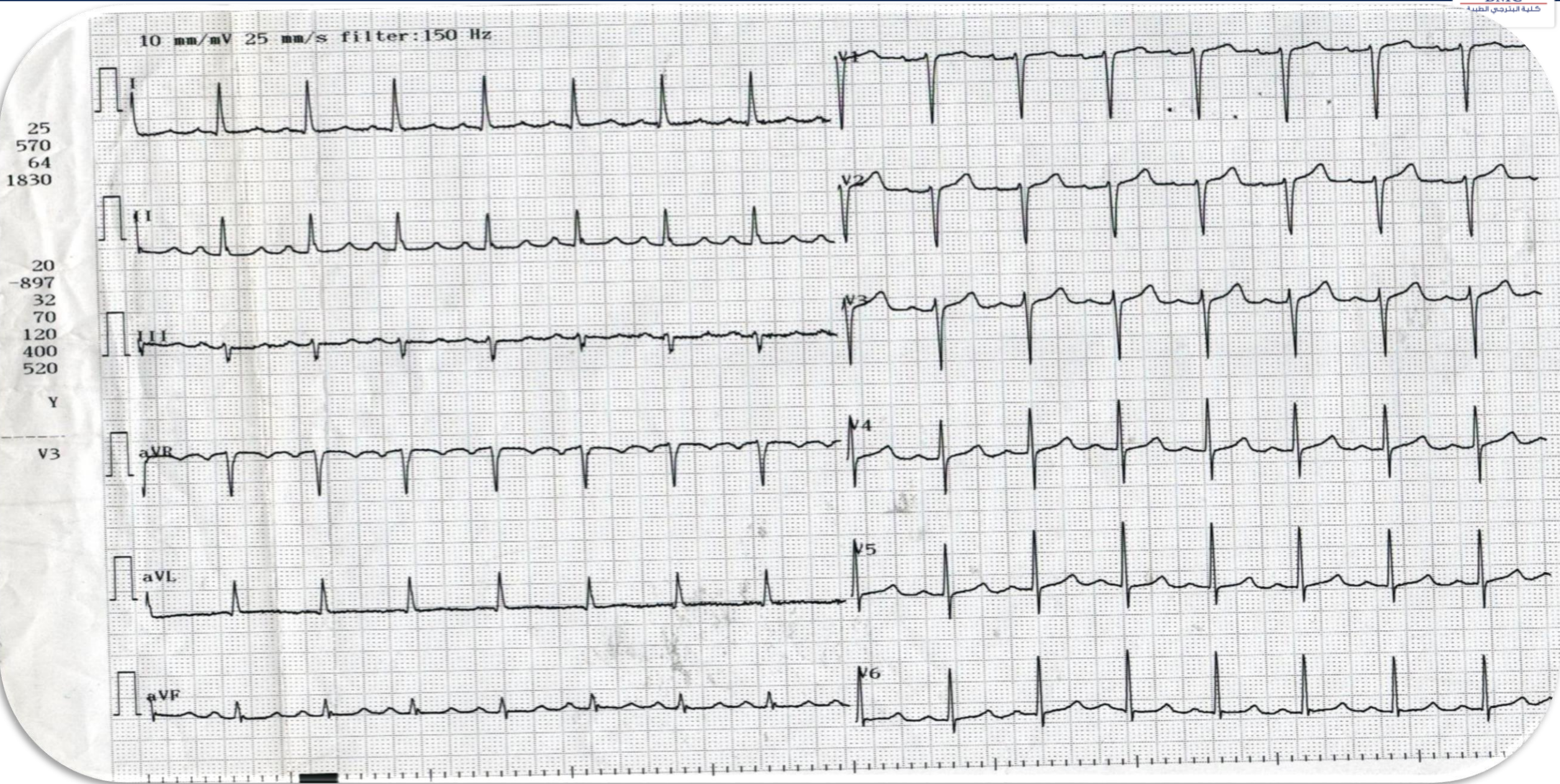
140-280 bpm

Narrow

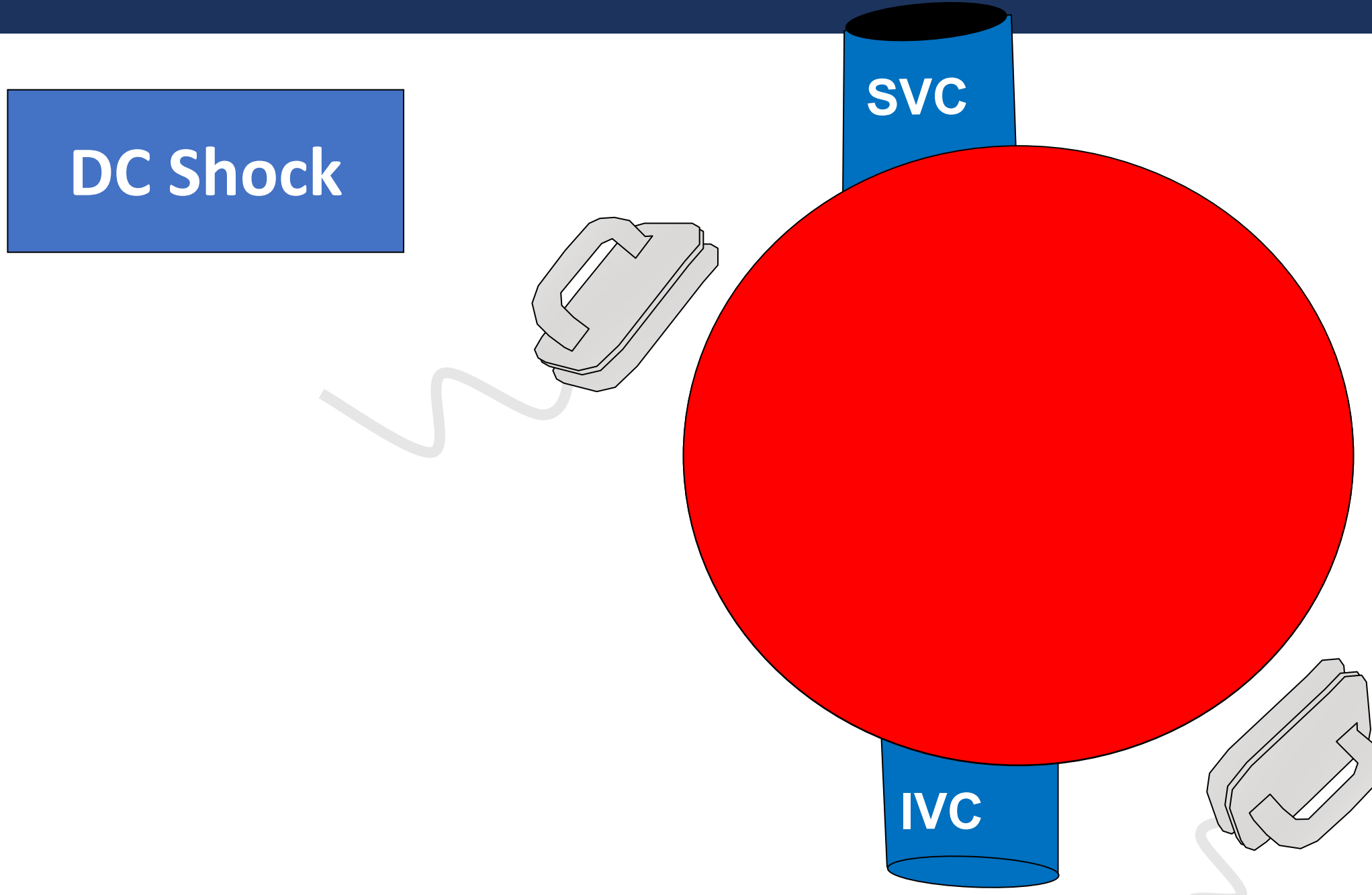
SVT



After cardioversion



DC Cardioversion

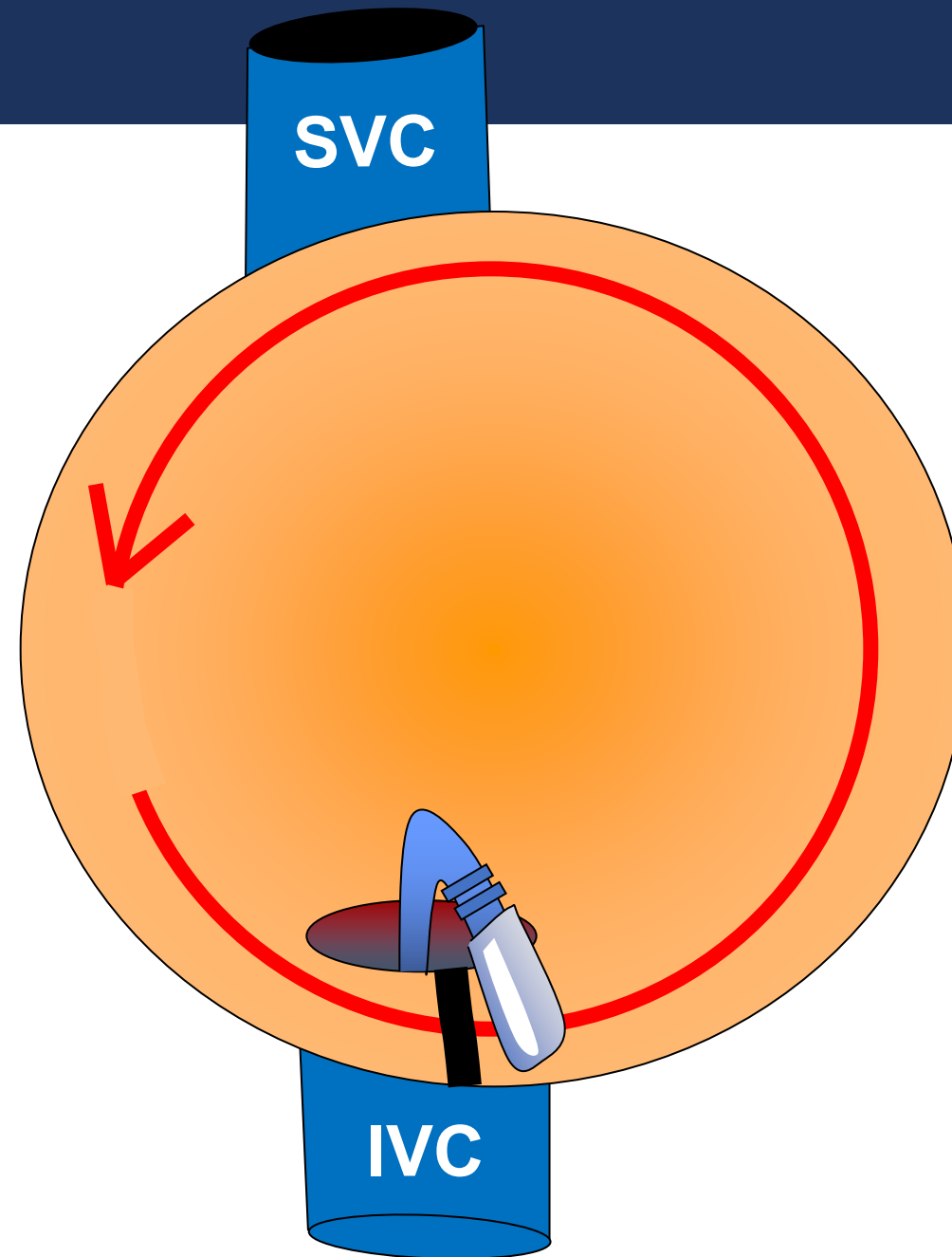


- Normal: maintenance, B-blocker/verapamil
- W-P-W: Transfer for EPS & RF ablation.

RF Ablation

RF ablation means destruction of tissue by radiofrequency current (heat $\sim 80^{\circ}\text{C}$).

RF Ablation



Narrow Irregular (A-fib)

- Causes: most commonly AF.
- A-fib classification

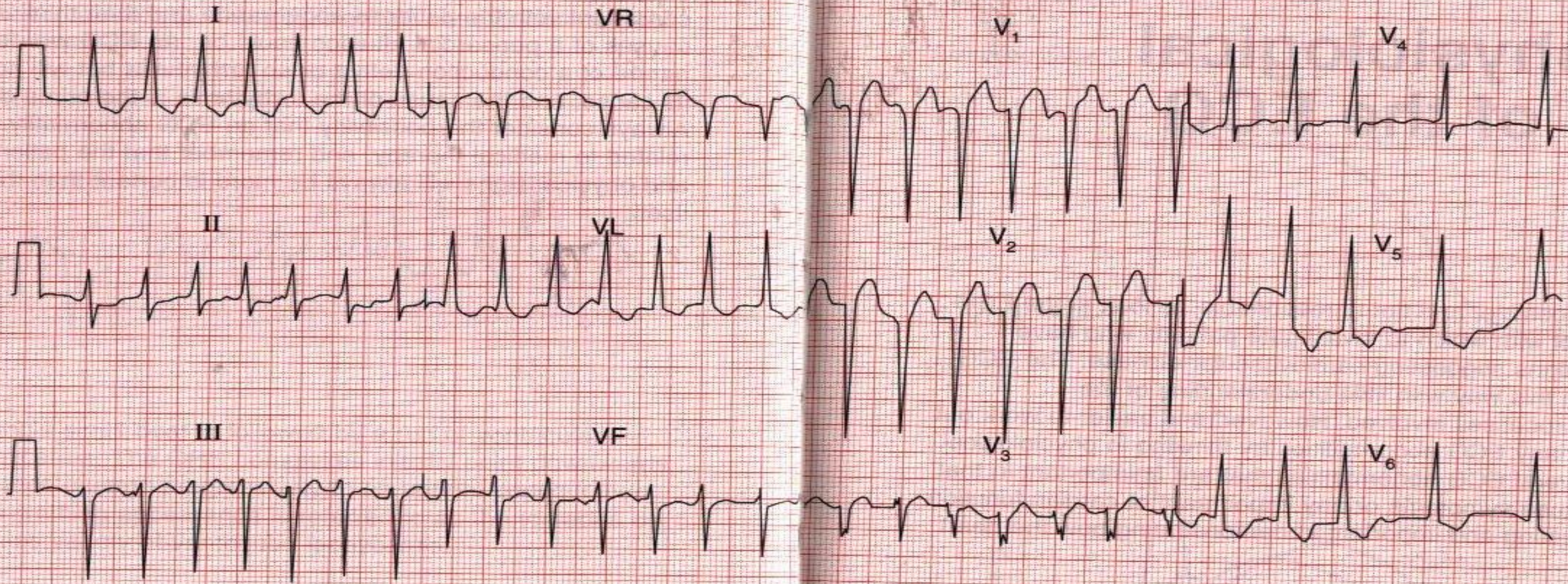
Ventricular rate	Afib with rapid ventricular response	Afib with a ventricular rate > 100 – 110 /minute (<u>tachycardic Afib</u>) ^{[10][11]}
	Afib with slow ventricular response ^[12]	Afib with a ventricular rate < 60 /minute (<u>bradycardic Afib</u> or <u>slow Afib</u>)
Onset and duration	New-onset Afib	New presentation or <u>diagnosis of Afib</u> , regardless of duration ^[13]
	Paroxysmal Afib	Afib that resolves within 7 days of onset either following treatment or spontaneously; the frequency of recurring episodes may vary.
	Persistent Afib	Continuous Afib for > 7 days
	Long-standing persistent Afib	Continuous Afib for > 1 year
	Permanent Afib	Persistent Afib in which therapeutic attempts are no longer made to convert to or maintain <u>sinus rhythm</u> unless the patient and the treating physician agree to do so ^[5]

- Hemodynamic unstable .. DC shock 150J.S
- Stable and of recent onset (i.e. < 48 hrs)

Medical reversion by :

- ❖ Oral propafenone (600 mg) **in absence** of CHF/impaired LV systolic function.
- ❖ IV amiodarone:
 - ✓ **Loading** 300 mg iv in 40 min.
 - ✓ **Infusion** of 1800 mg/24 hr

A-fib

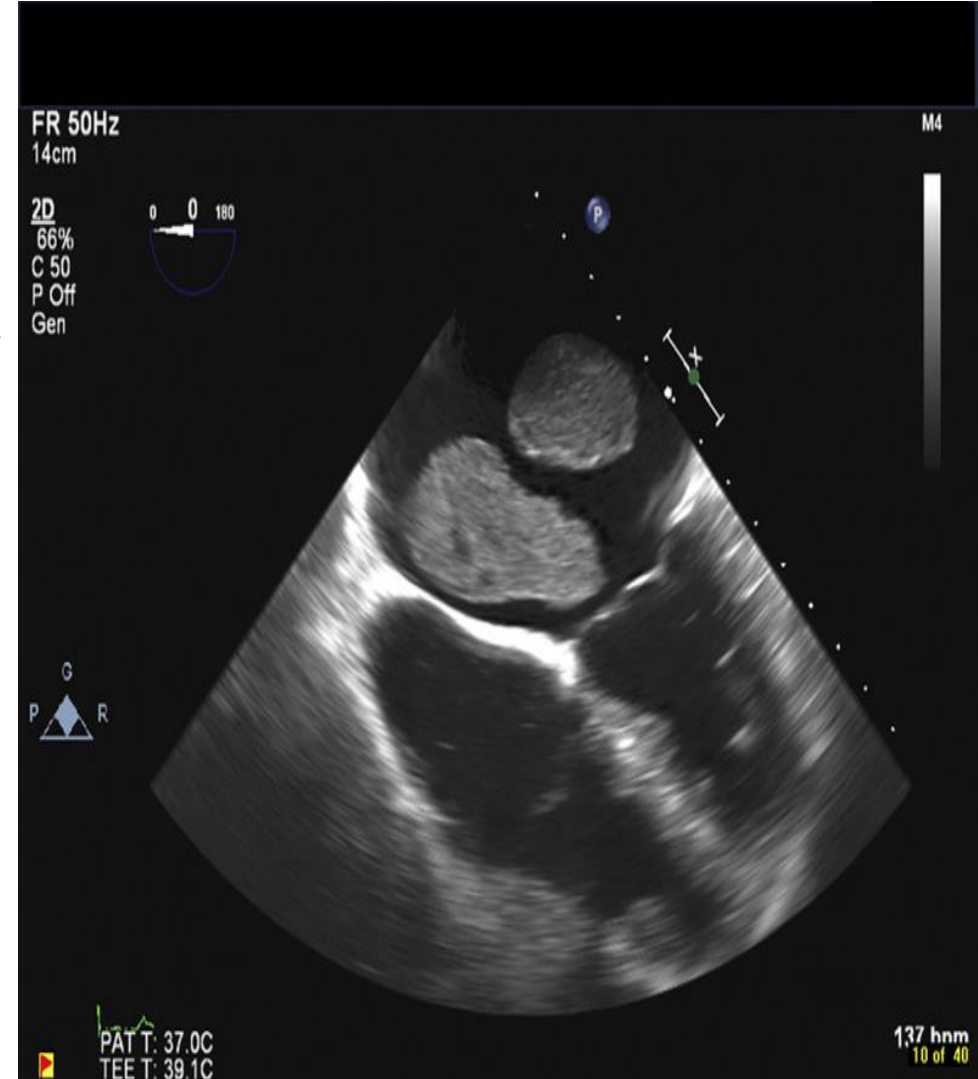


Lead II rhythm

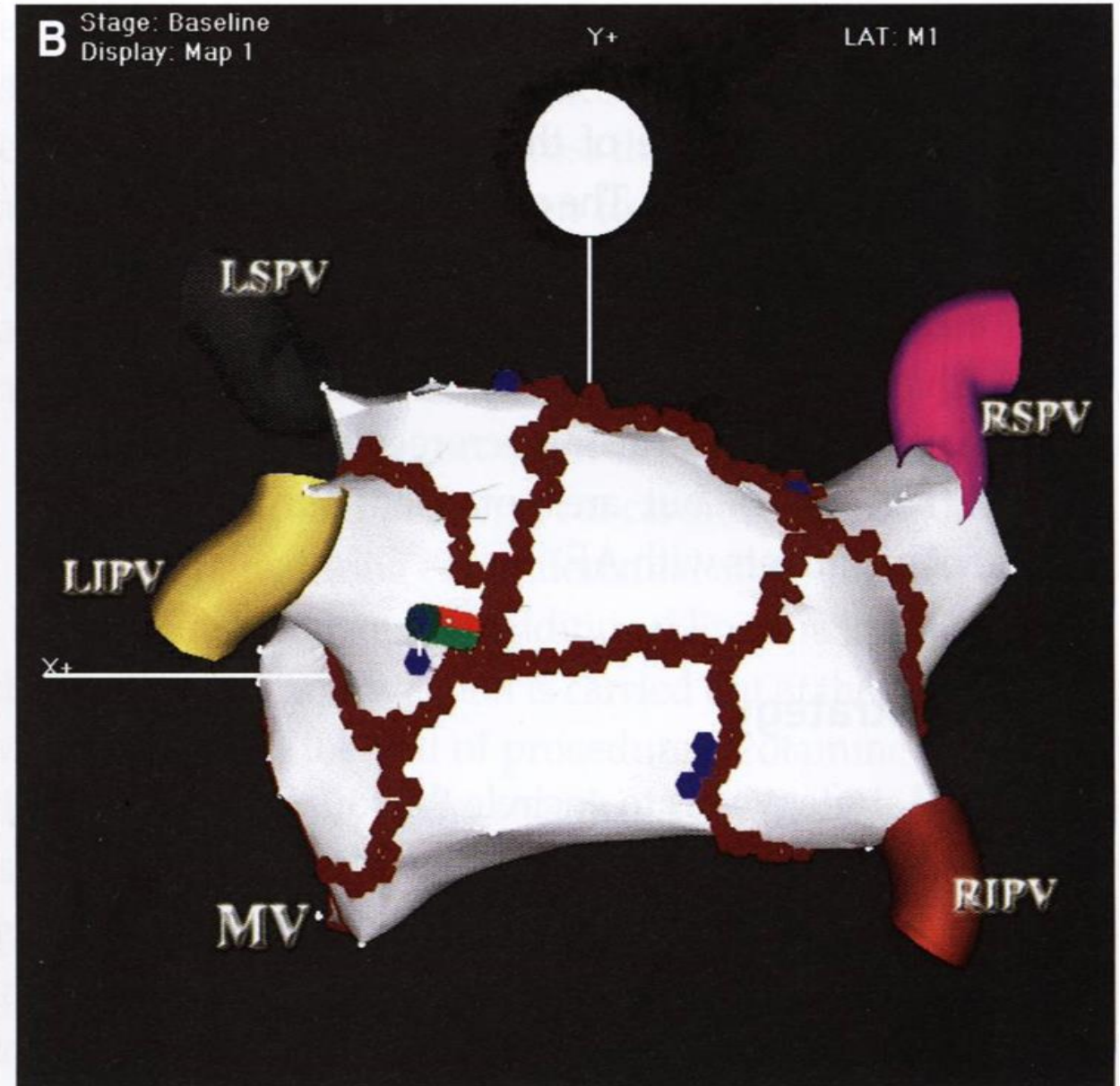
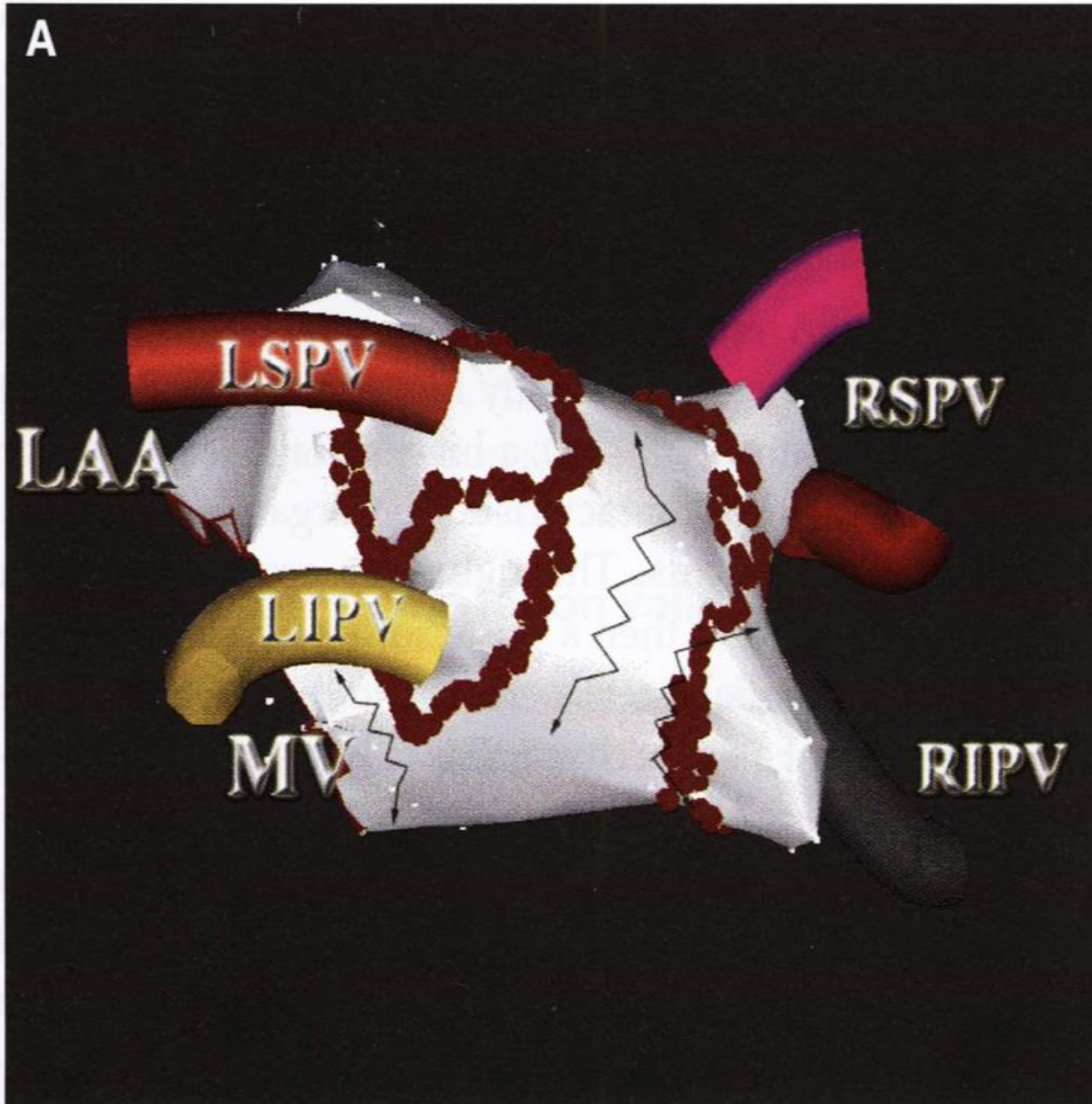


A-fib > 48 hrs

- Exclude LA thrombus by TEE.
- Coagulation before reversion.
- Prophylactic anticoagulation after reversion with Warfarin (INR 2-3).
- Maintenance after cardioversion:
 - ✓ Amiodarone 200 mg/day
 - ✓ Sotalol (80 mg Bid)
- Frequent recurrence → PV isolation.



Pulmonary vein isolation



Rate control for atrial fibrillation:

- Pharmacological options for rate control
 - First-line
 - ✓ Beta blockers
 - ✓ Non-dihydropyridine calcium channel blockers
 - Second line: digoxin
 - Third line: amiodarone
- Surgical options for rate control
 - AV nodal ablation and implantation of a permanent ventricular pacemaker
- Do not forget to anticoagulate .. INR 2-3

Afib -Risk factors

CHA₂DS₂-VASc score

Condition	CHADS ₂ score	Points	CHA ₂ DS ₂ -VASc score	Points
Congestive heart failure (or Left ventricular systolic dysfunction)	C	1	C	1
Hypertension: blood pressure consistently above 140/90 mmHg (or treated hypertension on medication)	H	1	H	1
Age ≥75 years	A	1	A ₂	2
Diabetes Mellitus	D	1	D	1
Stroke or TIA or thromboembolism in history	S ₂	2	S ₂	2
Vascular disease (e.g. peripheral artery disease, myocardial infarction, aortic plaque)			V	1
Age 65-74 years			A	1
Sex category (i.e. female gender)			Sc	1

This table shows the components of the CHADS₂ (Gage et al., JAMA 2001 [36]) and CHA₂DS₂-VASc scores (Lip et al., Chest 2010 [41]) tools to assess stroke risk in patients with AF. These risk assessment tools help to determine who should and who should not receive anticoagulation. CHA₂DS₂-VASc improves risk stratification in patients with CHADS₂=0 or 1, and allows for identification of patients at truly low risk.

Afib - Prevention of thromboembolic complications

High risk patient (Indications of long-term anticoagulation)

- Valvular disease
- hypertrophic cardiomyopathy, and/or
- CHA2DS2-VASc score ≥ 2 in men and ≥ 3 in women

Medium risk patient (Consider anticoagulation)

- Nonvalvular Afib with a CHA2DS2-VASc score of 1 in men or 2 in women

Low risk patient (Anticoagulation is not routinely recommended)

Nonvalvular Afib with a CHA2DS2-VASc score of 0 in men or 1 in women

The risk of bleeding due to anticoagulation should always be taken into consideration when initiating anticoagulation therapy

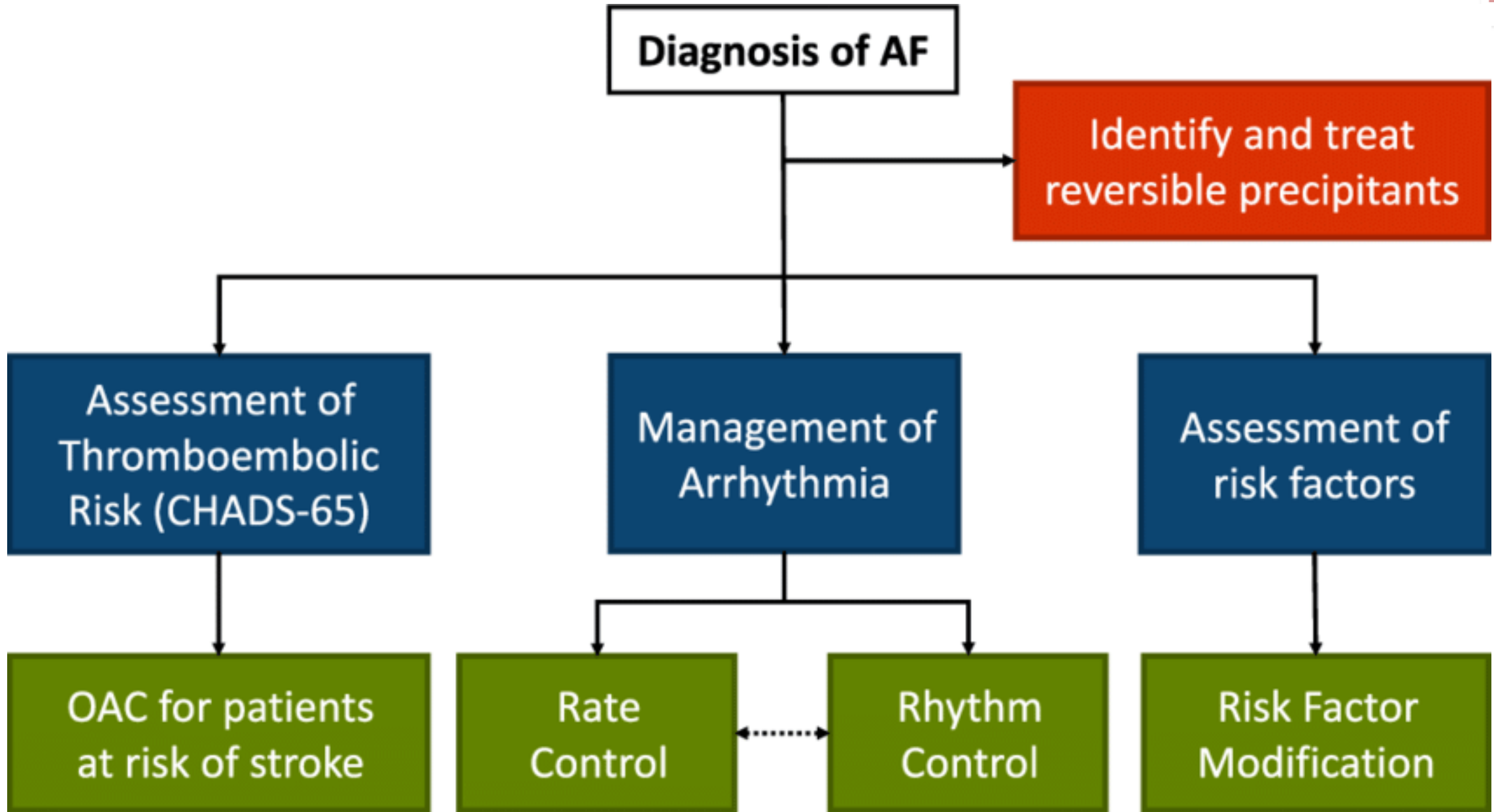
Afib - Prevention of thromboembolic complications

Clinical characteristic			Points
Hypertension	Uncontrolled, >160 mmHg systolic)	H	1
Abnormal renal and liver function (1 point each)	Dialysis, transplant, Cr >200 µmol/L, Cirrhosis, bilirubin >2x normal, AST/ALT/AP >3x normal)	A	1 or 2
Stroke history		S	1
Bleeding or predisposition to bleeding		B	1
Labile INR	Unstable/high INRs, time in therapeutic range < 60%)	L	1
Elderly	Age > 65	E	1
Drugs or alcohol (1 point each)	Antiplatelet agents, NSAIDs, ≥ 8 alcohol drinks/week	D	1 or 2

This table shows the components of the HAS-BLED score (Pisters et al, Chest 2010 [45], Camm et al., Eur H J 2010 [2]), used to assess bleeding risk.

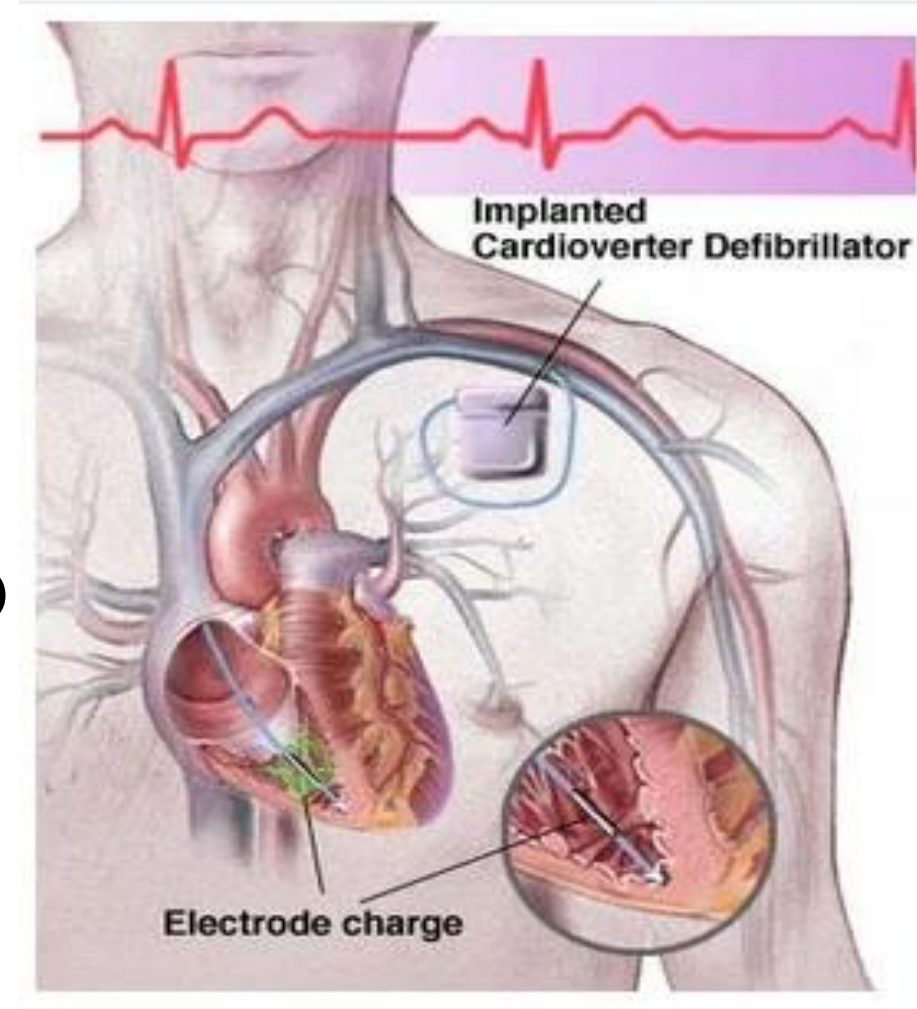
AF patients are subdivided into 3 risk stratifications,

- a score of 0 indicates low risk
- 1–2 indicates moderate risk
- ≥3 indicates high risk



Wide Regular Tachycardia

- Causes:
VT or SVT with aberration.
- VT Management
 - ✓ Unstable → DC shock 200 j.
 - ✓ Complicated acute MI → Xylocaine
 - ✓ Complicated IHD → Amiodarone + AICD
or RF ablation (low success rate)
 - ✓ Normal heart (young) → RF ablation



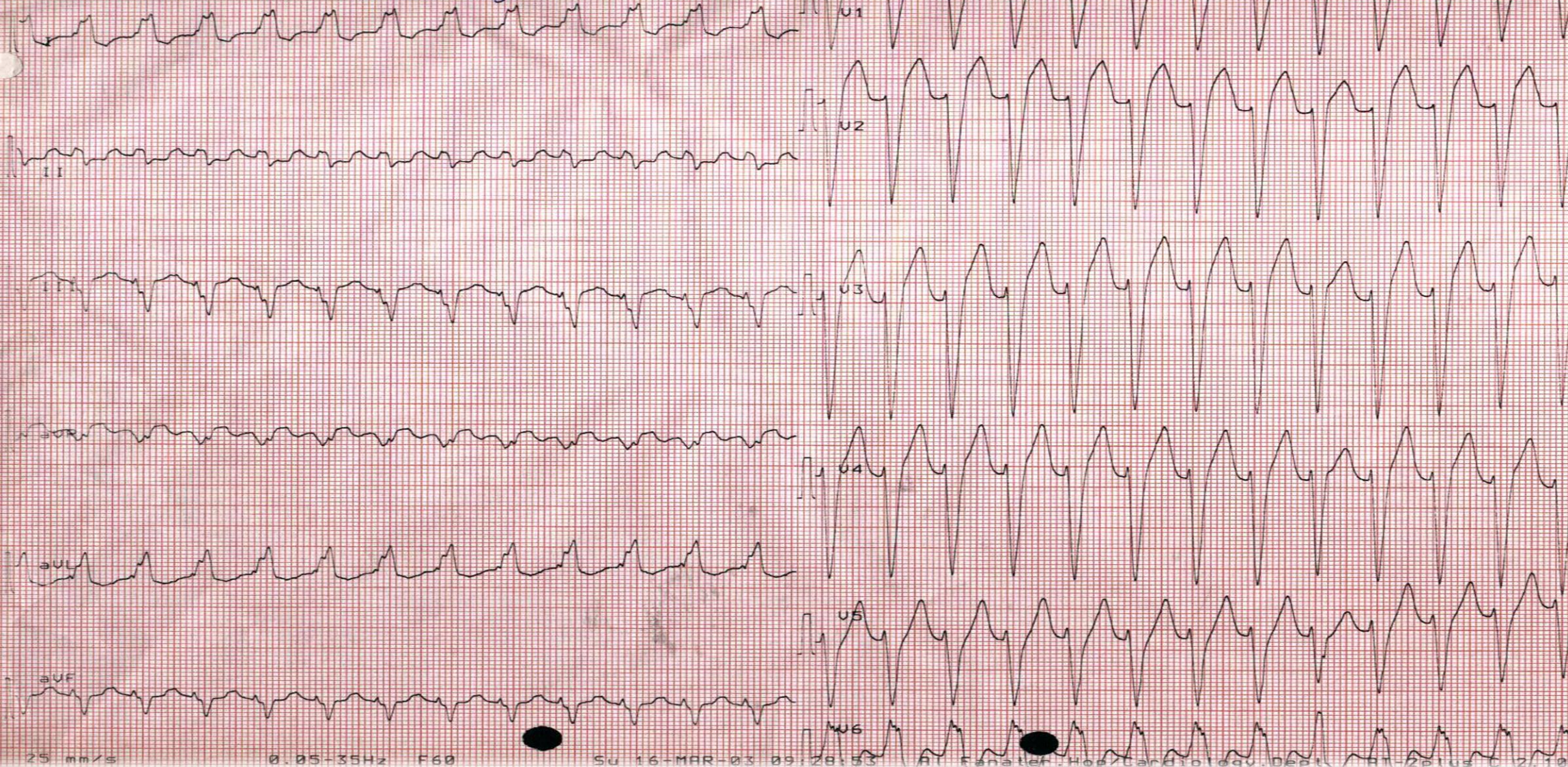
BP-120/90 mm Hg

10 mm/mV #357118
Abdulla Dazif

abdullah al dazif

10 mm/mV

HR 152/min



25 mm/s

0.05-35Hz F60

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Al Farateh Hosp Cardiology Dept RT-2plus C 2.10

Ventricular tachycardia (VT)

Definition

Ventricular tachycardia: ≥ 3 consecutive ventricular complexes (wide QRS complex) at a frequency of ≥ 100 /minute

Classification by duration

Nonsustained VT (NSVT): VT lasting < 30 seconds

Sustained VT : VT lasting ≥ 30 seconds

Classification by morphology

Monomorphic VT: QRS morphology similar in all beats, indicating a single arrhythmogenic focus

Polymorphic VT: QRS morphology varies in each beat, indicating multiple arrhythmogenic foci

Torsades de pointes (TdP)

Etiology

Congenital LQTS

Acquired LQTS (e.g., drug-induced LQTS, electrolyte abnormalities, heart disease)

Diagnostics

Characteristic ECG findings include:

Polymorphic VT with QRS complexes that typically appear to twist around the isoelectric line

Acute management

Hemodynamically unstable patients: Defibrillation plus CPR.

Hemodynamically stable patients: Administer IV magnesium sulfate.

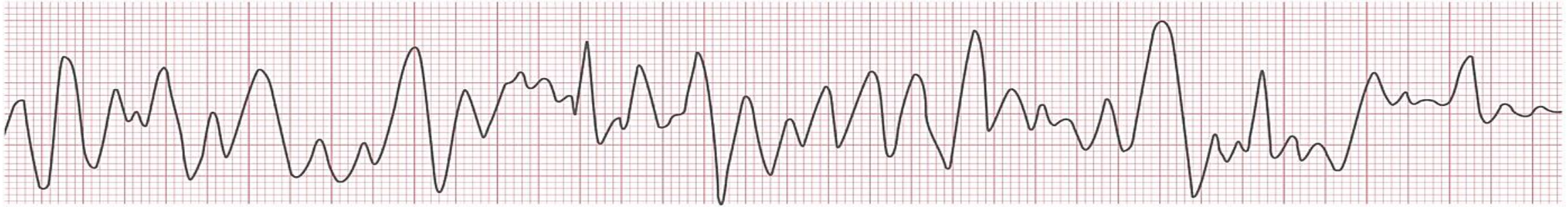
Complications: can progress to life-threatening ventricular fibrillation

Ventricular tachycardia

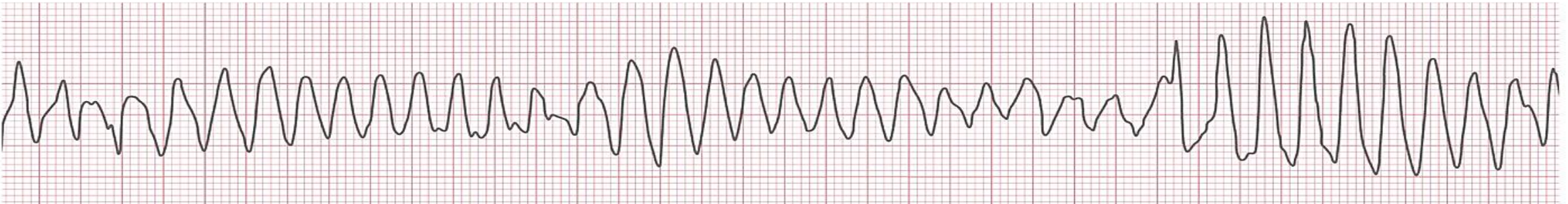
Monomorphic ventricular tachycardia



Polymorphic ventricular tachycardia

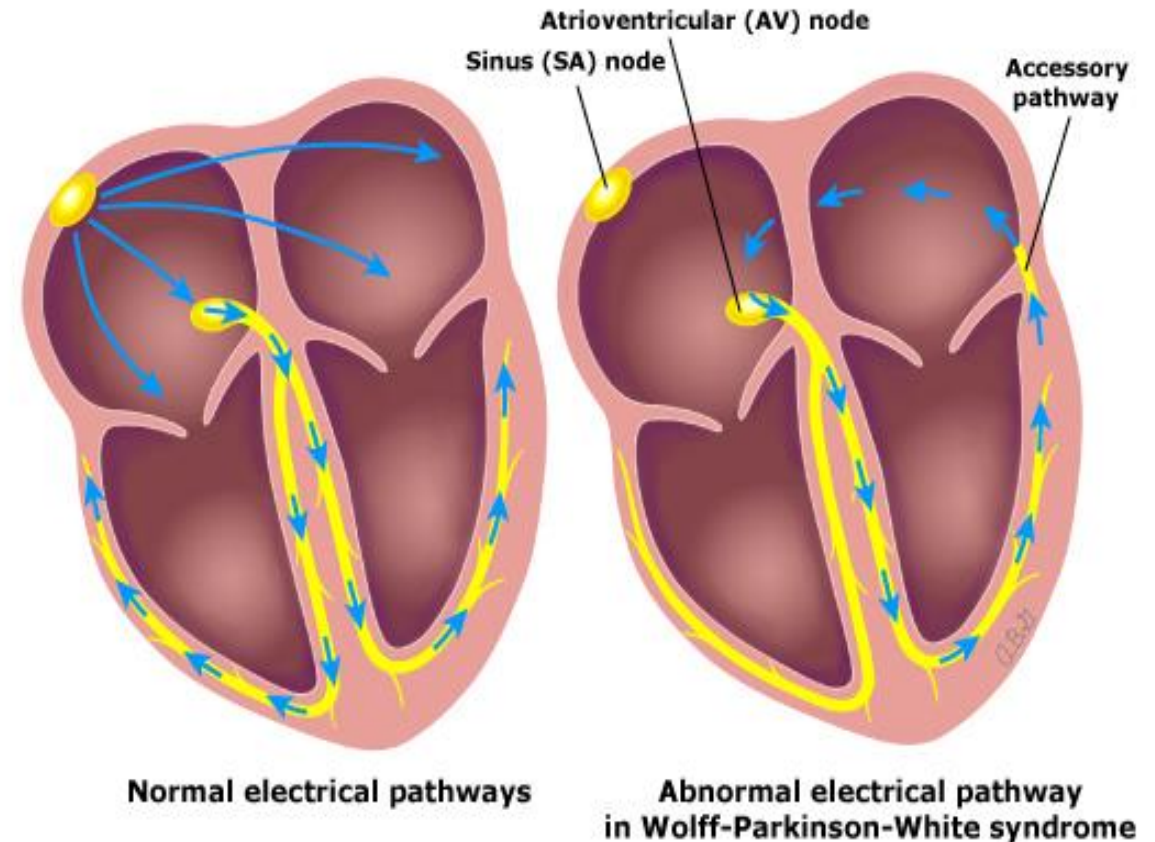


Torsades de pointes



Wide Irregular

- Causes: VF or AF complicating W-P-W
- Both are usually hemodynamic unstable, will require DC cardioversion.
- All drugs that block the AVN are contraindicated in treatment of AF complicating accessory pathway (WPW).

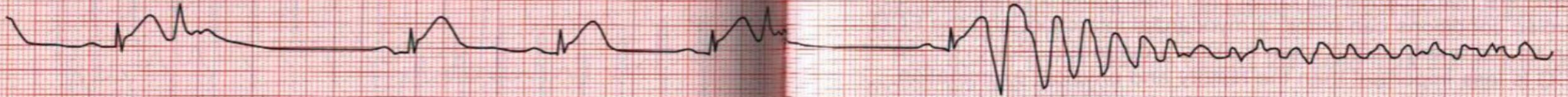


V- fib

(continuous record)



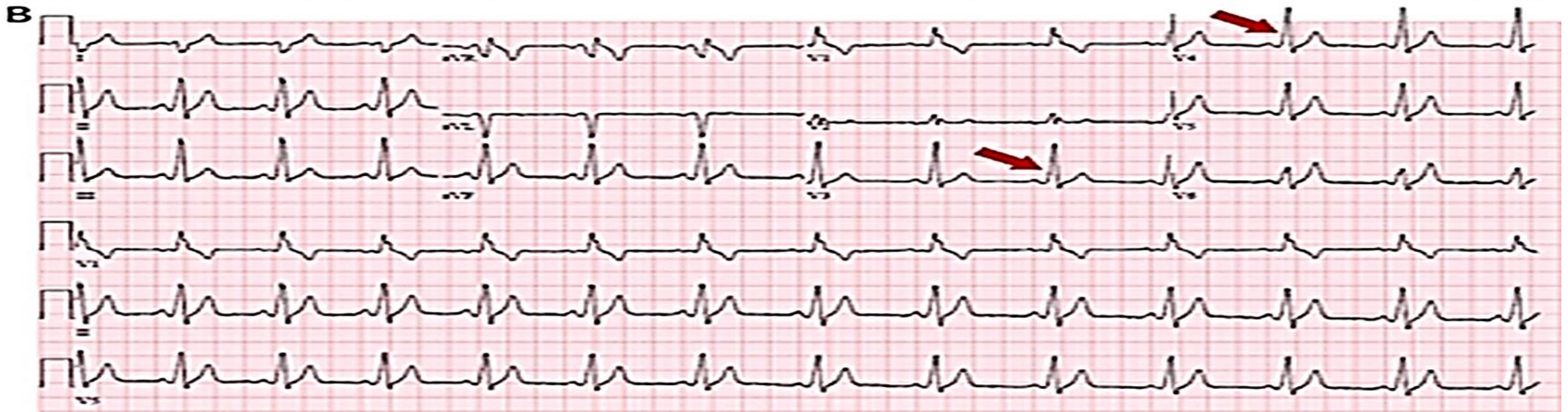
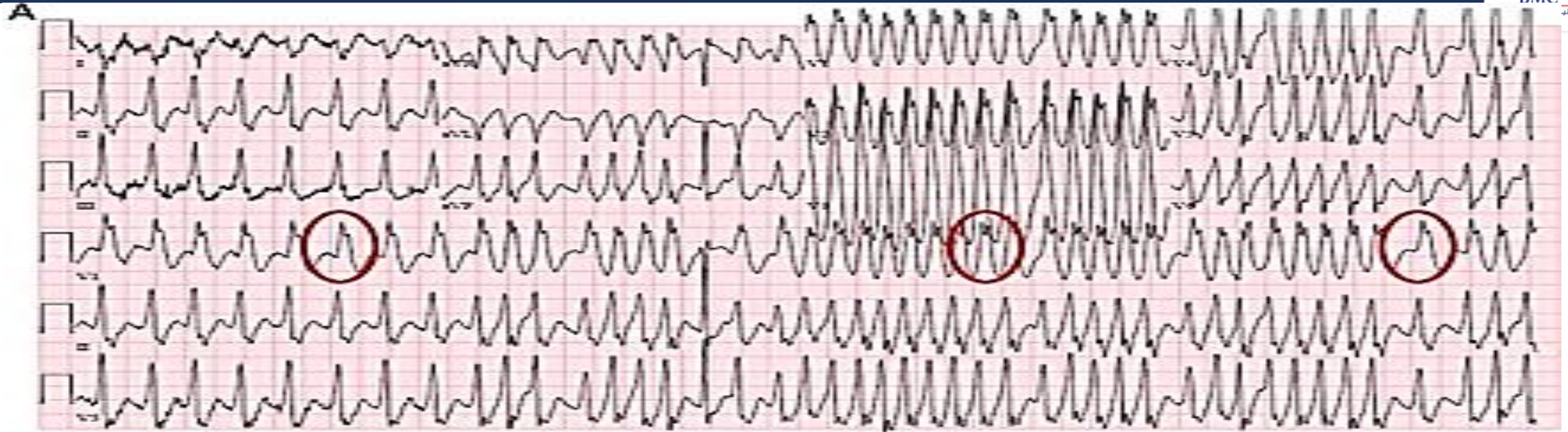
II (continuous record)



III (continuous record)



A-fib + WPW



Acute episodes

Unstable tachycardia : electrical cardioversion

Stable: Assess underlying rhythm.

Wide Complex Tachycardia (WCT):

irregular (e.g., underlying Afib or MAT) OR

regular (underlying Aflut or focal AT)

Avoid AV nodal blocking agents and vagal maneuvers

Rhythm control measures (i.e., cardioversion or IV procainamide) **are the safest treatment option**

Summary - Management of arrhythmias

- The vagal maneuver
- Pharmacological therapy (**anti-tachycardia agent & anti-bradycardia agent**).
- Cardioversion (electrical & chemical).
- Interventional therapy (Radiofrequency catheter ablation).
- Pacemaker therapy.

Vagal manoeuvre



The vagal maneuver:

- Techniques used to increase vagal parasympathetic tone in an attempt **to diagnose and treat** various arrhythmias.
- Done to abort episodes of stable supraventricular tachycardia (SVT) or differentiate SVT from ventricular tachycardias (VT).
- Vagal maneuvers include carotid sinus massage (CSM), and Valsalva maneuver (VM).

Vagal manoeuvre

Technique:

Carotid Sinus Massage – Place the patient in a supine position with their neck extended and rotated to the opposite side. The carotid sinus is at the arterial impulse just inferior to the angle of the mandible, anterior to the sternocleidomastoid, and the upper level of the thyroid cartilage.

- The provider uses the fingertips of their index and middle finger to apply pressure in the posteromedial direction for 5 to 10 seconds. If unsuccessful, the procedure is repeatable after 1 minute. If still unsuccessful, the clinician may attempt the maneuver on the opposite side.



Valsalva maneuver –Voluntarily increasing the intrathoracic pressure.

- The patient is placed in the supine position and asked to take a deep breath and blow out against a closed glottis for 10 to 15 seconds.



Management of arrhythmias (tachyarrhythmias)

Supraventricular Arrhythmias Only	Supraventricular and ventricular arrhythmias		Ventricular arrhythmias only
Adenosine	Amiodarone	Procainamide	Lidocaine
Verapamil	β blockers	Disopyramide	Mexilitine
Diltiazem	Propanolol	Quinidine	
Dronadarone	Sotalol	Flecainide	
Digoxin	Esmolol	Propafenone	

Table 21-1

Classification of Cardiac Antiarrhythmic Drugs

Class I (inhibit fast sodium ion channels)

Class IA

Quinidine
Procainamide
Disopyramide
Morcizine

Class IB

Lidocaine
Tocainide
Mexiletine

Class IC

Flecainide
Propafenone

Class II (decrease rate of depolarization)

Esmolol
Propranolol
Acebutolol

Class III (inhibit potassium ion channels)

Amiodarone
Sotalol
Ibutilide
Dofetilide
Bretylium

Class IV (inhibit slow calcium channels)

Verapamil
Diltiazem

Miscellaneous agents

Adenosine
Digoxin
Magnesium
Potassium

Proarrhythmia effect of antiarrhythmia agents

- Ia, Ic class: Prolong QT interval, will cause VT or VF in coronary artery disease and heart failure patients
- III class: Like Ia, Ic class agents
- II, IV class: Bradycardia

Management of arrhythmias

- **Cardioversion:**

Conversion to normal sinus rhythm.

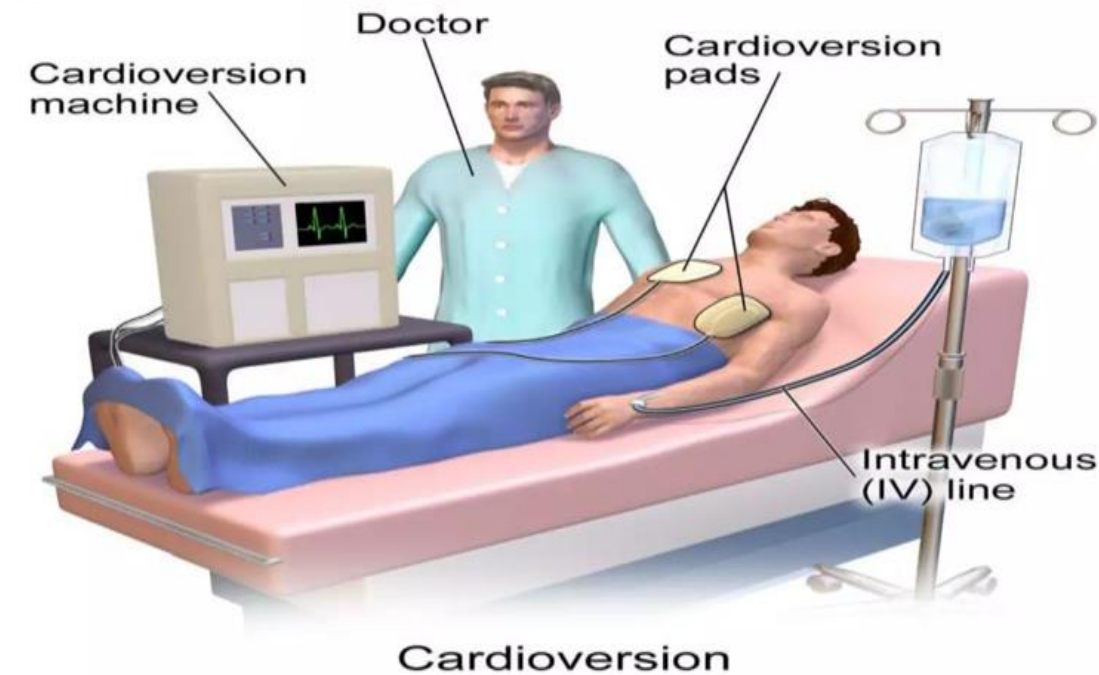
- *Two types of cardioversion:*

1- **Electrical cardioversion**, a high-energy shock is sent to the heart to reset a normal rhythm. (**synchronized Vs unsynchronized**)

(Defibrillation is a treatment for **life-threatening cardiac dysrhythmias**, especially ventricular fibrillation, and non-perfusing ventricular tachycardia.

- A heart that is in **asystole** (flat line in ECG) cannot be restarted by a **defibrillator** but would be **treated** by cardiopulmonary resuscitations (**CPR**).

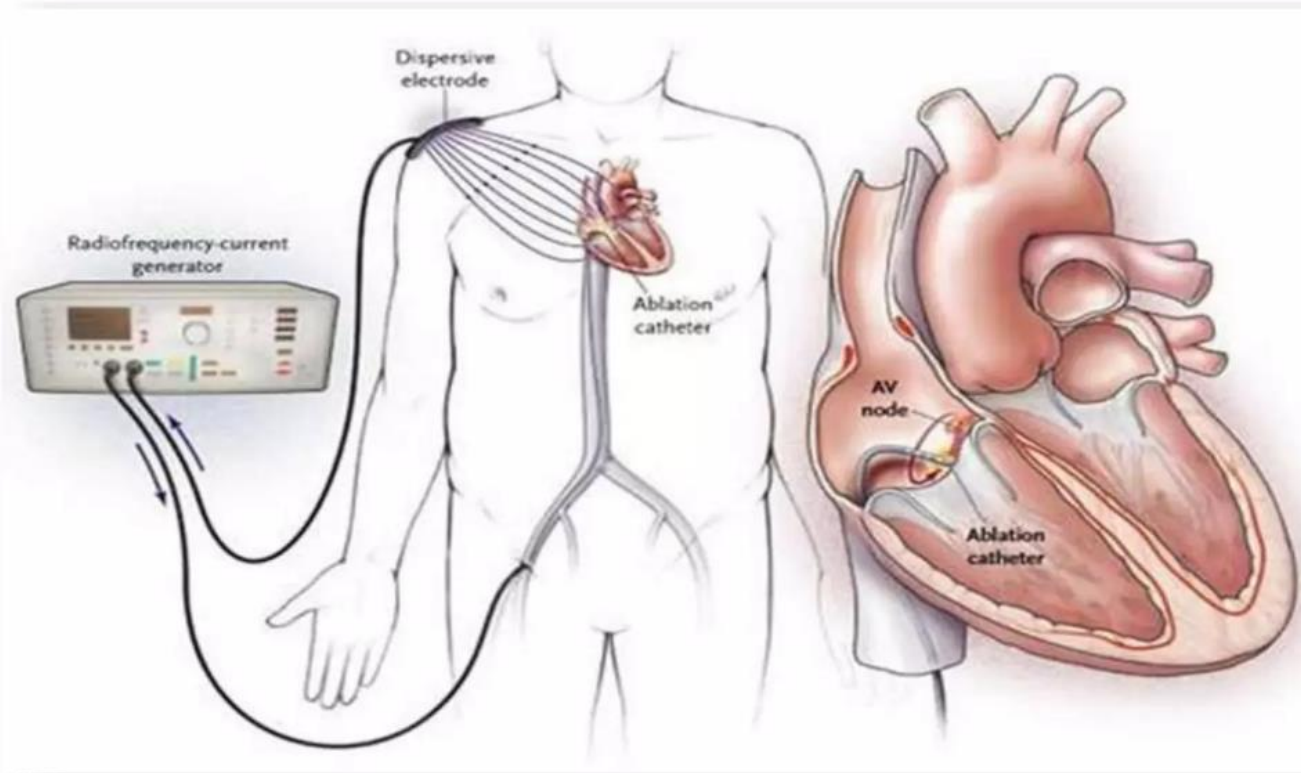
2- **Chemical cardioversion**, in which medicines are used to restore a normal rhythm.



Management of arrhythmias

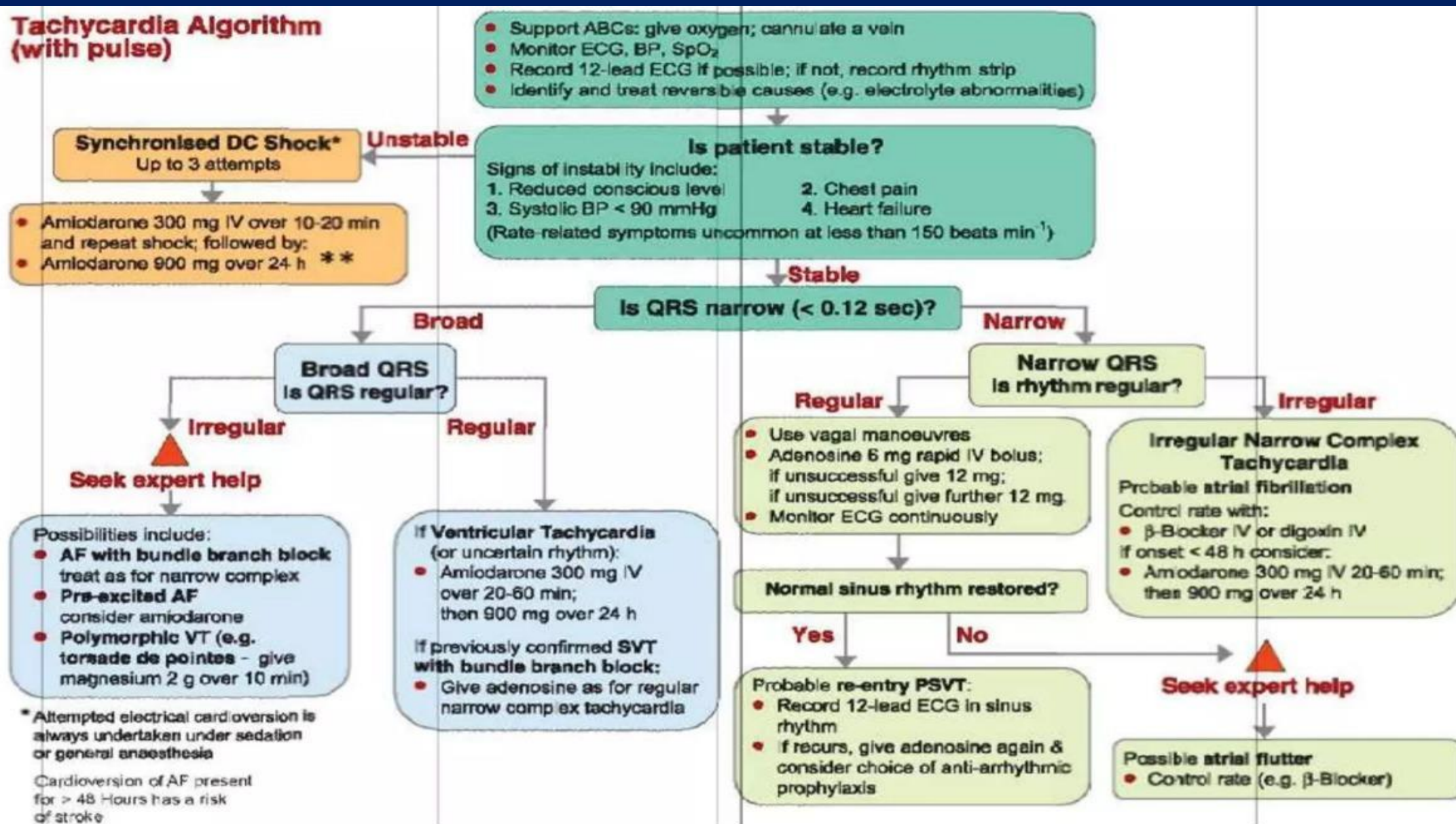
- Interventional therapy.

Radiofrequency catheter ablation



Summary Management of tachy-arrhythmias

Tachycardia Algorithm (with pulse)



References & further reading

- Al-Khatib SM, Stevenson WG, Ackerman MJ, et al. 2017 AHA/ACC/HRS Guideline for Management of Patients With Ventricular Arrhythmias and the Prevention of Sudden Cardiac Death. J Am Coll Cardiol. 2018; 72(14): p.e91-e220. doi: 10.1016/j.jacc.2017.10.054
- El-Sherif N, Turitto G, Boutjdir M. Acquired long QT syndrome and torsade de pointes. Pacing Clin Electrophysiol. 2018; 41(4): p.414-421. doi: 10.1111/pace.13296
- AMBOSS
- Harrison's Principles of Internal Medicine 20th edition ISBN-13: 978-1259644030

Thank You

Sometimes you need to change
direction!! Patience is not always
the optimal way

